

EVEWORLD: PHYSICAL EVOLUTION SUPERVISION FOR EMBODIED WORLD MODELS

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<https://behappy.github.io/EVEWorld/>

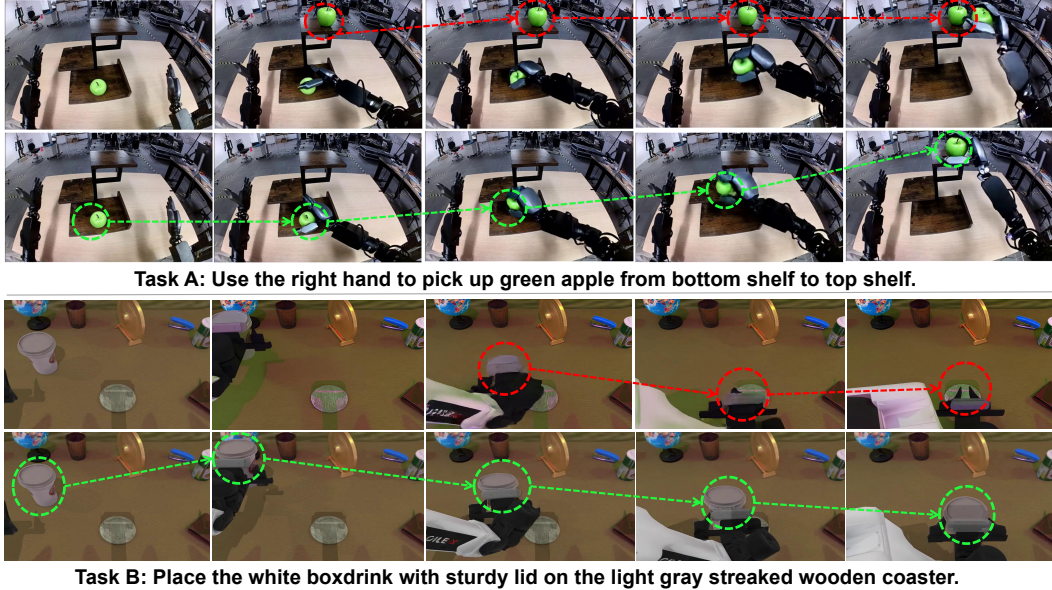


Figure 1: Embodied world models face the challenge of **Model Laziness**, where instance consistency is violated by spurious duplication or disappearance during interaction. **Red** denotes Standard SFT, which exhibits such failures, while **Green** denotes EVEWORLD, which better maintains physical consistency throughout the interaction. Dashed arrows trace target evolution over time.

ABSTRACT

Embodied world models enable scalable simulation of embodied interactions for robot learning. However, existing models are prone to **Model Laziness**, as they focus on visual fidelity at the expense of physical reasoning and lack process-level supervision over the temporal dynamics of manipulated objects. In this work, we propose EVEWORLD, a physical evolution-supervision framework for physically consistent target evolution. EVEWORLD consists of two components: Instance-Guided Restoration (IGR) and Temporal Instance Alignment (TIA). First, IGR promotes instance consistency through restoration supervision. Second, TIA promotes cross-frame consistency by aligning target instances across adjacent frames. We further introduce the Model Laziness Rate (MLR), a metric that measures persistent violations of instance consistency in generated trajectories. Extensive experiments on DreamGenBench, EWMBench, and PBench demonstrate the effectiveness of EVEWORLD, notably achieving an 87.5% reduction in MLR compared with GigaWorld-0. On the WorldArena 2.0 Track 1 leaderboard, our model ranks 6th in JEPA Similarity and 17th overall, which further validates the performance of our evolution supervision strategy.

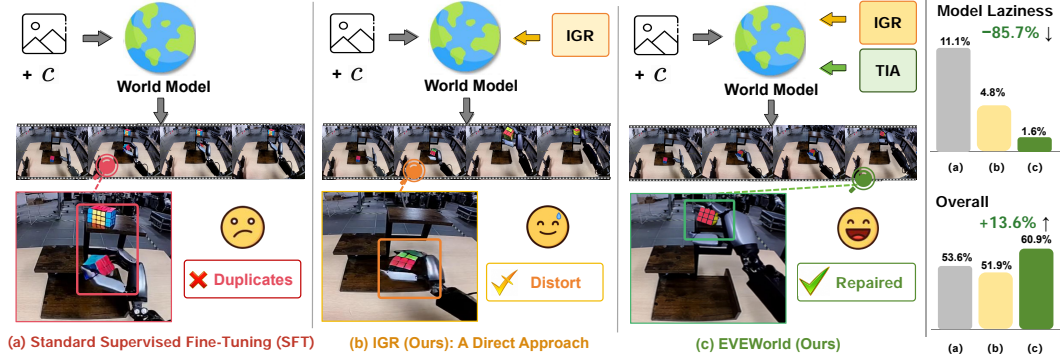


Figure 2: **Comparison of Standard SFT, IGR, and EVEWORLD.** (a) Existing video world models (GigaWorld Team et al., 2025; Yang et al., 2025; Team Wan et al., 2025; NVIDIA et al., 2025), including our Standard SFT baseline, may exhibit instance-consistency failures, such as target duplication at the goal. (b) Our IGR variant improves instance consistency, but cross-frame consistency failures such as target distortion may remain. (c) Our full EVEWORLD model combines IGR and TIA to maintain physically consistent target evolution throughout the interaction. Relative to Standard SFT, EVEWORLD reduces MLR by 85.7% and improves overall Gemini-IF by 13.6%. Additionally, the inputs consist of an initial image and an instruction c .

1 INTRODUCTION

Embodied world models and related policy-learning methods are emerging as scalable predictive engines for embodied intelligence, modeling diverse interaction dynamics to support robot learning, planning, and simulation (Yang et al., 2024a; Bruce et al., 2024; Zhou et al., 2024; Jang et al., 2025; Li et al., 2025b; Du et al., 2023; Schmidt & Jiang, 2024; Ye et al., 2025; Zitkovich et al., 2023; Kim et al., 2025; Black et al., 2025; Physical Intelligence et al., 2025). Recent work (Huang et al., 2025a; 2026; Guo et al., 2026; GigaWorld Team et al., 2026; Xiao et al., 2025a; Pang et al., 2025; Dou et al., 2026; Duan et al., 2026; Zheng et al., 2026; Bardhan et al., 2026; Li et al., 2025c; 2026; Quevedo et al., 2026) has further advanced embodied world modeling and its evaluation through high-fidelity generation, multiview and long-context modeling, action conditioning, persistent memory, and task-oriented evaluation.

Recent embodied world models further strengthen structural faithfulness through explicit 4D RGB–depth–normal prediction, geometry-aware action conditioning, and disentangled dynamics and visual synthesis, with related physical-evaluation benchmarks (Zhen et al., 2025; DreamX Team et al., 2026; Shan et al., 2026; Liu et al., 2026; Li et al., 2025a; 2025b; Bansal et al., 2026; Meng et al., 2025; Qin et al., 2025; Shang et al., 2026a; Motamed et al., 2026; Garrido et al., 2025). Nevertheless, stronger visual and structural modeling does not necessarily ensure physically consistent target evolution: a rollout may exhibit target-instance consistency failures, such as duplication or disappearance, and cross-frame consistency failures, such as deformation and identity drift (Yuan et al., 2026; Foo et al., 2026). Figure 1 shows a representative instance consistency failure after standard supervised fine-tuning (SFT). We refer to such instance consistency failures as **Model Laziness**: frame-level reconstruction can admit a shortcut that satisfies appearance or endpoint cues without maintaining instance consistency throughout the interaction.

In this work, we propose EVEWORLD, an evolution-supervision framework that promotes physically consistent target evolution and mitigates Model Laziness in embodied world models, as shown in Figure 2. EVEWORLD comprises two complementary components: **Instance-Guided Restoration (IGR)**, which promotes instance consistency by restoring count-perturbed videos, and **Temporal Instance Alignment (TIA)**, which promotes cross-frame consistency by aligning target instances across adjacent frames. To quantify Model Laziness, we introduce the **Model Laziness Rate (MLR)**, a metric that measures persistent violations of target-instance consistency over a rollout. Extensive evaluations on DreamGenBench, WorldArena 2.0, EWMBench, and PBench demonstrate the effectiveness of EVEWORLD in promoting physically consistent target evolution. Notably, on the WorldArena 2.0 Track 1 leaderboard, EVEWORLD achieves an EWMScore-P of 70.17, ranking 17th overall and 6th in JEPa Similarity. These results further support the effectiveness of evolution supervision under large-scale world-model evaluation.

2 RELATED WORK

Video Generation Models for Robotics. Large-scale video models provide strong priors for embodied prediction and scalable robot-data generation (Blattmann et al., 2023; Brooks et al., 2024; Team Wan et al., 2025; Kong et al., 2024), enabling world models for robot learning, planning, and simulation (Yang et al., 2024a; Zhou et al., 2024; Jang et al., 2025; Lu et al., 2025; Gao et al., 2025; GigaWorld Team et al., 2025). Recent work (Zhen et al., 2025; DreamX Team et al., 2026; Shan et al., 2026; Fu et al., 2026) further strengthens structural modeling through richer representations and conditioning: Tesseract jointly predicts RGB, depth, and normals, DreamX-Phi incorporates geometric action conditioning and depth supervision, and DVG-WM disentangles dynamics from visual synthesis, and RoboMaster improves multi-object interaction modeling. In contrast, our EVEWORLD retains the host model’s native representation and introduces evolution supervision during post-training to promote physically consistent target evolution.

Physical Reasoning in Embodied World Models. Prior works (Yang et al., 2024b; Traub et al., 2024; Jeong et al., 2025; Chen et al., 2025; Wu et al., 2026; Xiao et al., 2025a; Dou et al., 2026) improve object-centric and temporally consistent modeling through correspondence, tracking, object permanence, and persistent memory, while other studies (Deng et al., 2026; Sun et al., 2025; Khose et al., 2026) examine object dynamics, quantity behavior, and broader aspects of physical reasoning from different perspectives. However, despite these advances and evaluations (Yang et al., 2024b; Traub et al., 2024; Jeong et al., 2025; Xiao et al., 2025a; Dou et al., 2026; Deng et al., 2026; Sun et al., 2025; Khose et al., 2026), physically consistent target evolution remains challenging, with manipulated objects exhibiting failures in both instance consistency and cross-frame consistency. In contrast, our EVEWORLD introduces evolution supervision to explicitly promote physically consistent target evolution.

3 WHY EMBODIED WORLD MODELS BECOME LAZY

3.1 PROBLEM STATEMENT

Given an initial observation I_0 and an instruction c , an embodied video world model predicts future frames $x_{1:T}$. We focus on manipulation scenarios that require instance consistency, where the target-instance count is expected to remain unchanged throughout the interaction.

Let $N_t(I_0, x_t, c)$ denote the number of instances of the target specified by c , with $N_0(I_0, c)$ denoting its initial count. We define Model Laziness as

$$\text{ML}(I_0, x_{1:T}, c) = \mathbb{I}[\exists t \in \{1, \dots, T\} : N_t(I_0, x_t, c) \neq N_0(I_0, c)]. \quad (1)$$

This captures instance consistency failures caused by unsupported target duplication or disappearance. Interactions that intentionally change the target-instance count, such as cutting or assembly, are outside the scope of the current formulation and MLR protocol. Robot-induced occlusion is handled separately through an occlusion-aware check during MLR evaluation.

3.2 CAUSES OF MODEL LAZINESS

Standard training emphasizes visual reconstruction while providing limited process-level supervision for physically consistent target evolution (Denton & Fergus, 2018; Bengio et al., 2015; Jeong et al., 2025). This leaves two key gaps: instance consistency and cross-frame consistency.

Missing Instance-Consistency Supervision. Experimental results show that Standard SFT retains 92%–99% of the injected target area across different perturbation settings (Appendix Figure 10, Appendix C.2). Because the perturbed region is small relative to the full frame, its reconstruction error can be diluted by the surrounding scene, providing insufficient supervision for recovering the correct target-instance count. IGR (Section 4.1) addresses this gap by providing a direct restoration target and assigning greater spatial weight to the perturbed region.

Missing Cross-frame Consistency Supervision. IGR improves instance consistency, but a correct target-instance count alone does not ensure cross-frame consistency; the target may still become distorted, fade, or shift abruptly across adjacent frames, as shown in Figure 2(b). This stems from

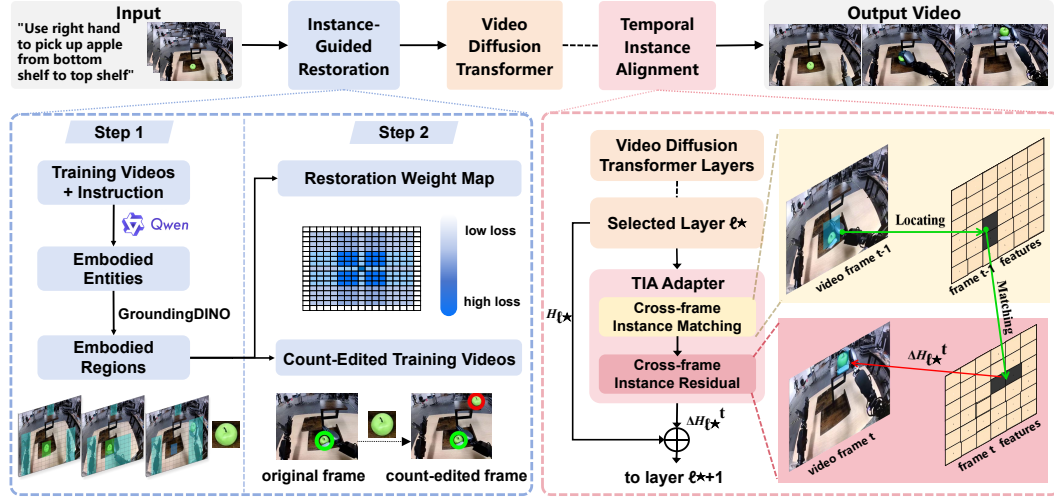


Figure 3: **Overall architecture of EVEWORLD.** Given a clean demonstration x and instruction c during training, IGR constructs a count-edited video $\tilde{x}$, which is processed by the video diffusion transformer to restore x . Within the transformer, TIA operates at the selected layer ℓ^* to perform cross-frame instance matching and residual refinement, after which the refined features pass through the remaining layers to produce the final output.

the lack of explicit supervision linking the same target across frames, allowing locally plausible predictions to evolve inconsistently over time. TIA (Section 4.2) addresses this gap by aligning target instances across adjacent frames.

4 METHOD

Framework Overview. Given a clean demonstration video x and its instruction c , EVEWORLD uses IGR (Section 4.1) to construct a count-edited video $\tilde{x}$ and trains the model to restore x , assigning higher weights to restoration-related regions to promote instance consistency. TIA (Section 4.2) operates within the video diffusion transformer to align target features across adjacent frames and promote cross-frame consistency. Together, IGR and TIA enable physically consistent target evolution in the final video output. Notably, the demonstration video is used for training supervision; at inference, EVEWORLD takes only the initial image and instruction as input.

4.1 INSTANCE-GUIDED RESTORATION

To address failures in physically consistent target evolution, we introduce Instance-Guided Restoration (IGR) to promote instance consistency. IGR inserts an extra target instance into clean demonstrations to form count-edited samples, then trains the model to restore the corresponding clean latent representation encoded by a pretrained VAE. A language model and GroundingDINO (Liu et al., 2024a) localize the target trajectory to sample a target patch p and valid paste region Ω :

$$\tilde{x} = \text{Paste}(x, p, \Omega), \quad z = \text{VAE}(x), \quad \tilde{z} = \text{VAE}(\tilde{x}). \quad (2)$$

Training the model to map the count-edited latent $\tilde{z}$ back to the clean latent z encourages recovery of the correct target-instance count. To prevent this local instance-consistency signal from being diluted by global reconstruction, we assign restoration-related regions three times the reconstruction weight of ordinary regions and normalize the resulting spatial weight map to unit mean. We denote the normalized map by $\hat{W}$; its exact construction is given in Appendix B.1. Using the EDM-style noise weighting $\lambda(\sigma)$ (Karras et al., 2022), the restoration objective is

$$\mathcal{L}_{\text{IGR}} = \mathbb{E} \left[\lambda(\sigma) \frac{1}{CTHW} \left| \hat{W}^{1/2} \odot (D_{\theta}(\tilde{z} + \sigma \epsilon, c, \sigma) - z) \right|_2^2 \right]. \quad (3)$$

Minimizing $\mathcal{L}_{\text{IGR}}$ provides explicit instance-consistency supervision by recovering the correct target-instance state from count-perturbed inputs; details are provided in Appendix B.1.

4.2 TEMPORAL INSTANCE ALIGNMENT

IGR improves instance consistency, but a correct target-instance count alone does not guarantee cross-frame consistency. Failures such as identity drift, distortion, fading, or abrupt motion may still occur across adjacent frames. We therefore introduce Temporal Instance Alignment (TIA) to explicitly promote cross-frame consistency by aligning target instances across adjacent frames.

We select the attachment layer using an offline layer-wise matching probe. For each candidate layer ℓ , the frame- $(t-1)$ target feature retrieves its best match over all token cells of frame t , and we select the layer with the lowest endpoint error (EPE):

$$\hat{u}_t^\ell = \arg \max_v f_{t-1}^\ell(u_{t-1})^\top f_t^\ell(v), \quad \ell^* = \arg \min_{\ell \in \mathcal{B}} \frac{1}{|\mathcal{V}|} \sum_{t \in \mathcal{V}} \|\hat{u}_t^\ell - u_t\|_2. \quad (4)$$

where $\mathcal{V}$ contains adjacent-frame pairs with the target visible in both frames and detector confidence above a threshold. At layer ℓ^* , TIA has two stages (Figure 3, right). *Local instance locating* matches target features within a local neighborhood, while *cross-frame instance transport* converts pre-transport cosine similarities into softmax transport weights, splats projected previous-frame features into frame t , normalizes by transport mass, and writes them back through a zero-initialized bounded residual.

$$H'_t = H_t^{\ell^*} + \gamma \tanh\left(W_{\text{out}}(\bar{H}_t - W_{\text{in}}H_t^{\ell^*})\right). \quad (5)$$

Here, $\bar{H}_t$ denotes the transported feature field. GroundingDINO and target locations u_t provide offline alignment supervision during training only; at inference, dense matching and transport operate directly on the model’s hidden features and require neither target localization nor an external tracker. The transported features are further supervised to favor the true next-frame target location over nearby candidates:

$$\mathcal{L}_{\text{TIA}} = -\frac{1}{|\mathcal{V}|} \sum_{t \in \mathcal{V}} \log \frac{\exp(s_{t,u_t}/\tau)}{\sum_{v \in \mathcal{C}_t} \exp(s_{t,v}/\tau)}, \quad s_{t,v} = \bar{f}_{t-1}^{\ell^*}(u_{t-1})^\top \bar{f}_t^{\ell^*}(v), \quad (6)$$

where $\bar{f}^{\ell^*}$ is the normalized feature after transport. The locating, transport, and alignment components jointly constrain target identity across frames. Further details are provided in Appendix B.2.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP AND METRICS

Benchmarks and Data Splits. We use the official DreamGen GR1 fine-tuning split, which contains 92 robot-manipulation videos paired with instructions, and primarily evaluate on DreamGenBench, comprising 126 manipulation tasks across unseen environments, objects, and behaviors (Jang et al., 2025). We further evaluate cross-domain generalization on WorldArena 1.0, cross-trial robustness on PBench, cross-distribution generalization on AgiBot-EWMBench, and cross-backbone generalization using FlowWAM on WorldArena 2.0 and held-out RoboTwin actions (Shang et al., 2026a;b; AgiBot-World-Contributors et al., 2025; Hu et al., 2025; Mu et al., 2025; NVIDIA, 2025b; Huang et al., 2024; Liu et al., 2024b; Unterthiner et al., 2018; Chen et al., 2026a).

Evaluation Metrics. We evaluate instruction following with Qwen-IF and Gemini-IF (Bai et al., 2025; Gemini Team et al., 2023); WorldArena 1.0/2.0, EWMBench, PBench, and RoboTwin assess complementary aspects of perceptual quality, motion, semantics, consistency, and task fidelity.

Related evaluations (Deng et al., 2026; Khose et al., 2026; Sun et al., 2025), including RBench, WFM-Eval, and video numeracy benchmarks, provide indirect signals related to object quantity and persistence. In contrast, we propose the **Model Laziness Rate (MLR)**, a metric that explicitly measures target-instance consistency at the rollout level using the observed initial count N_0 as its reference, without requiring a prompt-specified count or reference future. We exclude under-counts attributable to robot-induced occlusion while retaining all over-counts, and denote the resulting

Table 1: **Overall comparison on DreamGenBench.** MLR denotes Model Laziness Rate, and Qwen-IF/Gemini-IF measure instruction-following accuracy. General video models are evaluated zero-shot, whereas Standard SFT and EVEWORLD are post-trained on DreamGen. MLR is computed on a shared set of 63 eligible items across all models.

Metric	CogVideoX	Wan-TI2V	Wan-A14B	Cosmos	GigaWorld-0	Standard SFT	EVEWorld
MLR (%) ↓	28.57	17.46	11.11	14.29	12.70	11.11	1.59
Qwen-IF (%) ↑	38.89	38.89	64.29	62.70	79.37	73.81	80.16
Gemini-IF (%) ↑	5.56	10.32	15.87	24.60	60.19	53.57	60.85

Table 2: **Fine-grained comparison on DreamGenBench.** GR1-Env, GR1-Object, and GR1-Behavior denote generalization to unseen environments, objects, and behaviors, respectively. Qwen-IF and Gemini-IF measure instruction-following accuracy. General video models are evaluated zero-shot, whereas Standard SFT and EVEWORLD share the same DreamGen post-training setting.

Method	Param.	GR1-Env IF (%)		GR1-Object IF (%)		GR1-Behavior IF (%)	
		Qwen ↑	Gemini ↑	Qwen ↑	Gemini ↑	Qwen ↑	Gemini ↑
<i>General video models (zero-shot)</i>							
CogVideoX1.5-5B-I2V	5B	48.28	15.52	24.00	1.00	48.94	4.26
Wan2.2-TI2V-5B	5B	51.72	17.24	24.00	5.00	46.81	11.70
Wan2.2-I2V-A14B	A14B	58.62	20.69	66.00	13.00	65.96	15.96
Cosmos-Predict2-2B	2B	68.97	31.03	60.00	28.00	61.70	17.02
<i>GigaWorld-based models</i>							
GigaWorld-0	2B	79.31	54.60	82.00	54.33	76.60	69.86
Standard SFT	2B	79.31	51.72	76.00	42.00	68.09	67.02
EVEWorld	2B	82.76	72.41	82.00	50.33	76.60	64.89

evaluation count by $\tilde{N}_t(x)$. MLR implements Eq. 1 with a two-frame persistence criterion to suppress isolated detection errors. We define

$$\text{MLR} = \frac{100}{|\mathcal{D}_+|} \sum_{x \in \mathcal{D}_+} \mathbb{I} \left[\exists t \in \{1, \dots, T-1\} : \tilde{N}_t(x) \neq N_0(x) \wedge \tilde{N}_{t+1}(x) \neq N_0(x) \right], \quad (7)$$

where $\mathcal{D}_+ = \{x \in \mathcal{D} : N_0(x) > 0\}$. Further details are provided in Appendix A.

Model and Training Setup. For the main experiments, we initialize EVEWORLD from the 2B-parameter GigaWorld-0 Video-Pretrain-2B (GigaWorld Team et al., 2025) and compare against CogVideoX1.5, Wan2.2, and Cosmos-Predict2 as zero-shot baselines (Yang et al., 2025; Team Wan et al., 2025; NVIDIA, 2025a). Standard SFT, EVEWORLD, and all ablations share the same initialization, training data, and 250-step optimization budget. We fine-tune all transformer parameters on eight NVIDIA H20Z GPUs with an effective batch size of 64 and a learning rate of 4.32×10^{-5} . For cross-distribution and cross-backbone experiments, we apply the same evolution-supervision method to the corresponding data and backbone. Details are provided in the appendix.

5.2 MAIN RESULTS

Overall Results. As shown in Table 1, EVEWORLD substantially mitigates Model Laziness while improving instruction following. Compared with Standard SFT, EVEWORLD reduces MLR from 11.11% to 1.59% (85.7% relative reduction), while Qwen-IF increases from 73.81% to 80.16% and Gemini-IF from 53.57% to 60.85%. Appendix A (Appendix Figure 8) further shows that the remaining violations under EVEWORLD are transient, indicating that evolution supervision improves physical consistency without sacrificing instruction following.

Fine-grained Analysis. Table 2 reports instruction-following performance across the Environment, Object, and Behavior splits. Compared with Standard SFT, EVEWORLD improves Qwen-IF on all three splits, with the largest gain on Behavior (68.09% to 76.60%), while Gemini-IF improves notably on Environment (51.72% to 72.41%) and Object (42.00% to 50.33%). Overall, these results show that evolution supervision yields broad gains across different generalization dimensions.

Table 3: **Zero-shot comparison on WorldArena 1.0.** All metrics follow the released evaluation protocol. IQ, AQ, DD, Flow, MS, SC, BC, and PC denote Image Quality, Aesthetic Quality, Dynamic Degree, Flow Score, Motion Smoothness, Subject Consistency, Background Consistency, and Photometric Consistency; Overall is their equal-weight mean. MLR is computed on 157 shared eligible items, and bold marks the best GigaWorld-based result.

Method	Appearance		Motion			Consistency			Main Metrics	
	IQ $\uparrow$	AQ $\uparrow$	DD $\uparrow$	Flow $\uparrow$	MS $\uparrow$	SC $\uparrow$	BC $\uparrow$	PC $\uparrow$	Overall $\uparrow$	MLR (%) $\downarrow$
<i>General I2V models (zero-shot)</i>										
CogVideoX	24.51	34.80	20.71	7.24	50.78	54.96	66.35	10.20	33.69	2.55
Wan-TI2V	33.48	37.90	33.00	11.38	61.86	68.88	77.17	18.19	42.73	17.20
Wan-A14B	51.93	43.37	42.29	8.04	64.13	67.69	75.37	33.61	48.30	29.94
Cosmos	57.88	38.83	50.46	16.69	71.61	64.14	73.71	21.42	49.34	8.28
<i>GigaWorld-based models</i>										
GigaWorld-0	58.53	38.79	44.36	7.74	61.67	77.90	88.13	48.32	53.18	27.39
Standard SFT	60.71	38.36	48.49	7.82	64.04	79.34	88.67	44.17	53.95	27.39
EVEWorld	59.82	37.35	62.90	10.98	67.82	78.72	88.91	47.58	56.76	13.38

Table 4: **Comparison on EWMBench after AgiBot post-training.** Standard SFT and EVEWORLD use the same AgiBot data and optimization budget. Motion, Semantics, and Overall are the official EWMBench aggregate scores, and DYN, HSD, nDTW, and Scene are component metrics. Complete results are provided in Appendix E.1.

Method	Motion $\uparrow$	Semantics $\uparrow$	DYN $\uparrow$	HSD $\uparrow$	nDTW $\uparrow$	Scene $\uparrow$	Overall $\uparrow$
GigaWorld-0	50.30	2.2497	7.19	23.29	19.83	89.58	3.6486
Standard SFT	61.51	2.2477	14.88	24.35	22.28	84.38	3.7066
EVEWorld	63.65	2.2524	17.55	24.12	21.97	86.37	3.7525

Table 5: **Cross-backbone evaluation and ablation on RoboTwin.** PSNR, SSIM, and LPIPS measure frame-level visual fidelity against aligned RGB ground truth, and Flow-EPE denotes optical-flow endpoint error.

Variant	IGR	TIA	PSNR (dB) $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Flow-EPE $\downarrow$	MLR (%) $\downarrow$
FlowWAM			12.218	0.748	0.383	3.033	52.05
Standard SFT			11.687	0.735	0.414	2.828	47.89
+ IGR	$\checkmark$		12.544	0.770	0.380	2.487	40.03
+ TIA		$\checkmark$	13.445	0.776	0.333	1.850	22.54
EVEWorld	$\checkmark$	$\checkmark$	12.765	0.769	0.365	2.207	35.21

5.3 GENERALIZATION EVALUATION

Zero-Shot Domain Generalization. On WorldArena 1.0 (Shang et al., 2026a), which includes unseen scenes and robot embodiments, EVEWORLD improves EWMScore-local-8 from 53.95 to 56.76 (+5.21%) over Standard SFT while reducing MLR from 27.39% to 13.38% (−51.15%) (Table 3). These results demonstrate improved cross-domain generalization in both motion quality and target-instance consistency.

Generalization Across Training Distributions. On EWMBench (Hu et al., 2025), EVEWORLD trained on the same AgiBot data (AgiBot-World-Contributors et al., 2025) and optimization budget as Standard SFT improves Motion and Overall by 3.48% and 1.24%, respectively (Table 4). These results show that the gains of EVEWORLD extend beyond the DreamGen training distribution, particularly in motion modeling.

Cross-Backbone Generalization. We replace the GigaWorld image-text-conditioned IT2V backbone with FlowWAM’s flow-conditioned WAM architecture (Table 5). On FlowWAM, IGR improves all reported metrics over Standard SFT on held-out RoboTwin actions, but TIA yields consistently larger gains. The full model continues to outperform Standard SFT, suggesting that the contributions of IGR and TIA are backbone-dependent. The resulting FlowWAM-based EVEWorld model, Super-

Table 6: **Component ablation on DreamGenBench.** All variants share the same initialization, training data, and optimization budget. Gemini-IF measures instruction-following accuracy.

Variant	Component		Gemini-IF (%)				MLR (%) ↓
	IGR	TIA	Env ↑	Object ↑	Behavior ↑	Overall ↑	
Standard SFT			51.72	42.00	67.02	53.57	11.11
IGR only	✓		49.43	36.67	69.50	51.85	4.76
TIA only		✓	55.17	42.67	68.79	55.29	7.94
EVEWorld	✓	✓	72.41	50.33	64.89	60.85	1.59

Table 7: **Exploration of Alternative Approaches.** PhysicsLatent uses weakly supervised process tokens; EAG uses energy-guided denoising; LAD-LoRA uses energy-regularized LoRA; Causal Frontier uses causal block-wise generation; ICH-D uses learned duplicate removal; and EVEWORLD combines instance restoration with temporal alignment. Each route is evaluated against its matched control; for EVEWORLD, the matched control is Standard SFT. Dyn., Train, Test, Causal, and Repair denote explicit dynamics/evolution modeling, training-time intervention, test-time intervention, causal generation, and learned repair, respectively. Δ denotes relative change from the matched control. Further details are provided in Appendix D.

Route	Design elements					Relative change	
	Dyn.	Train	Test	Causal	Repair	Δ MLR (%) ↓	Δ Gemini-IF (%) ↑
PhysicsLatent	✓	✓				↓ 7.0	↓ 11.9
EAG sampling guidance	✓		✓			↑ 150.0	↑ 10.0
LAD-LoRA	✓	✓				0.0	↓ 6.1
Causal Frontier		✓		✓		↑ 35.4	↓ 53.8
ICH-D	✓	✓	✓		✓	↑ 16.6	↓ 30.6
EVEWORLD (ours)	✓	✓			✓	↓ 85.7	↑ 13.6

Table 8: **Cross-trial robustness on PBench.** Each SFT₂₅₀–EVEWORLD pair uses the same inference seed, and the three trials use different seeds under otherwise identical settings. Scores denote the overall PBench Robot score, judged by Gemini-3.8-Flash. Δ denotes the gain over SFT₂₅₀, and Mean Δ averages the three paired gains.

Horizon	Trial 1			Trial 2			Trial 3			Mean Δ
	SFT ₂₅₀	EVEWORLD	Δ	SFT ₂₅₀	EVEWORLD	Δ	SFT ₂₅₀	EVEWORLD	Δ	
3.8 s	57.37	59.27	+1.90	53.40	55.07	+1.67	59.57	60.61	+1.04	+1.54
5.8 s	59.42	60.02	+0.60	60.36	63.02	+2.66	60.79	62.21	+1.42	+1.56

visionWM, achieves an EWMScore-P of 70.17 on the official WorldArena 2.0 Track 1 leaderboard, ranking 17th overall and 6th in JEPA Similarity. These results support the cross-backbone generalization of our evolution-supervision framework, while the optimal component configuration may depend on the model architecture.

5.4 ABLATION STUDIES AND ANALYSIS

Component Contributions of IGR and TIA. Table 6 shows that IGR primarily improves instance consistency, reducing MLR from 11.11% to 4.76%; TIA improves cross-frame consistency and Overall Gemini-IF. Combining both yields 1.59% MLR and a Gemini-IF score of 60.85, demonstrating complementary effects under the GigaWorld-0 image-text-conditioned IT2V architecture.

Cross-Trial Robustness. We further evaluate EVEWORLD on PBench Robot across multiple stochastic inference trials under a matched-budget setting, with evolution supervision as the only methodological difference from SFT₂₅₀. As shown in Table 8, EVEWORLD improves all six paired trials across 3.8 s and 5.8 s rollouts, with mean gains of +1.54 and +1.56, respectively, demonstrating consistent robustness across inference seeds and rollout horizons.

Exploration of Alternative Approaches. We explored five alternative approaches to mitigate Model Laziness while improving instruction fidelity: PhysicsLatent, EAG, LAD-LoRA, Causal

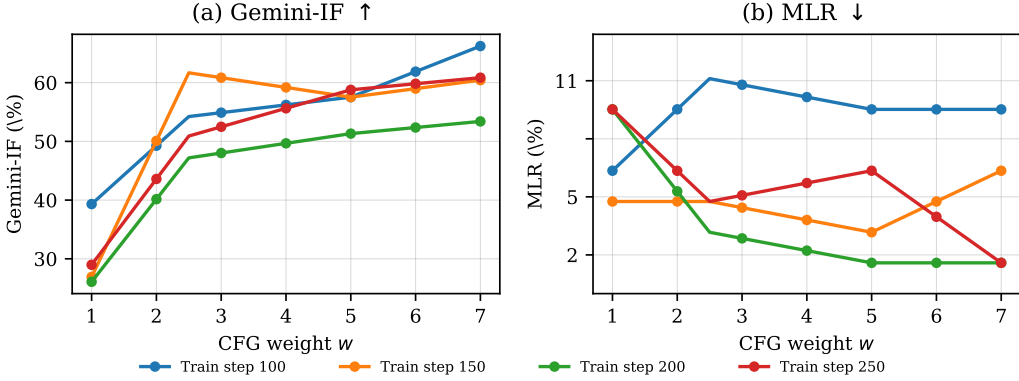


Figure 4: **CFG sensitivity across training stages on DreamGenBench.** Gemini-IF and MLR are evaluated under different classifier-free guidance weights w , with each curve denoting a training checkpoint.

Table 9: **Sequence-length scaling and inference cost.** We report latency (p50/p95: median/95th percentile), throughput, memory, and GPU cost across output durations.

Output		Latency (s) $\downarrow$		Throughput	Memory	Cost
Dur. (s)	Frames	p50	p95	Frames/min $\uparrow$	Peak/GPU (GiB) $\downarrow$	GPU-h/1k $\downarrow$
3.8	61	8.166	8.403	442.1	38.19	18.40
5.8	93	10.932	11.352	502.2	38.33	24.69
9.8	157	18.323	19.249	504.8	38.99	41.46
15.8	253	31.503	33.523	469.5	41.47	71.85
19.8	317	41.737	45.007	443.3	40.70	95.34

Frontier, and ICH-D (Table 7). However, none achieved both objectives simultaneously. For example, PhysicsLatent reduces Model Laziness at an 11.9% cost in instruction fidelity, whereas EAG gains 10.0% in instruction fidelity at a 150.0% increase in Model Laziness. We share these trade-offs to provide practical insights for future research, with further details in Appendix D.

CFG Sensitivity Analysis. Figure 4 shows that Gemini-IF generally increases with higher CFG weights, but MLR varies non-monotonically across guidance weights and training stages. Thus, CFG does not reliably improve instruction fidelity and reduce Model Laziness simultaneously, indicating the importance of explicit supervision for target evolution.

Sequence-Length Scaling and Inference Cost. Table 9 shows that increasing output duration from 3.8 to 19.8 s raises p50/p95 latency from 8.17/8.40 to 41.74/45.01 s and GPU-hour cost per 1,000 videos from 18.40 to 95.34, with relatively stable throughput and peak memory. Thus, longer rollouts mainly increase computation with limited additional memory overhead.

6 CONCLUSION

We presented EVEWORLD, a physical evolution supervision framework for embodied world models, and proposed the Model Laziness Rate (MLR), a metric to quantify target-instance consistency failures, which we term Model Laziness. EVEWORLD combines IGR for instance consistency and TIA for cross-frame consistency to promote physically consistent target evolution. Experiments show an 85.7% MLR reduction on DreamGenBench with improved instruction following, 17th overall and 6th in JEPA Similarity on the WorldArena 2.0 Track 1 leaderboard, and generalization gains across training distributions and model backbones.

Limitations and Future Works. The limitations of our current formulation mainly arise from its local, pairwise supervision, which does not explicitly capture long-horizon state dependencies or complex multi-object interactions. Future work will further expand the EVEWorld framework with broader forms of evolution supervision for longer-horizon and relational dynamics.

REPRODUCIBILITY STATEMENT

The appendices document the training and evaluation settings. The supplementary material and the public repository provide code, data indices, configurations, evaluation prompts, and item-level results. The repository is available at <https://github.com/behappya/EVEWorld>.

AI USE STATEMENT

During the preparation of this manuscript, the authors used generative AI tools solely for writing assistance, including language polishing, grammar checking, and improving clarity of presentation. The AI tools were not used for developing the method, conducting experiments, generating results, or making scientific decisions. All AI-assisted edits were reviewed and verified by the authors, who take full responsibility for the final content of this manuscript.

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APPENDIX OVERVIEW

This supplementary material provides additional details and results for EVEWORLD, organized as follows:

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A MODEL LAZINESS AND MLR

MLR detects persistent deviations from the initial target-instance count N_0 , while under-counts supported by robot-induced occlusion are exempted and all over-counts are retained. Appendix C.2 tests recovery from an injected count error. Appendix A.1 details the evaluation protocol and occlusion check; Appendices A.4–A.5 cover model, detector, and domain variation.

A.1 MLR: COMPARISON WITH EXISTING EVALUATION METRICS

Existing evaluations expose signals related to object persistence and quantity, but differ in both their evaluation anchors and aggregation units.

RBench. RBench evaluates spontaneous entity emergence and disappearance as part of Physical-Semantic Plausibility (PSS), together with other physical-semantic violations such as floating, penetration, and non-contact attachment (Deng et al., 2026). After normalization, PSS is combined with

Task-Adherence Consistency (TAC) into the Task Completion score:

$$\text{TC}_{\text{RBench}} = \frac{\text{PSS} + \text{TAC}}{2}. \quad (8)$$

Thus, appearance and disappearance failures contribute to a broader assessment of physical plausibility and task correctness rather than forming an independent rollout-level target-instance consistency metric. In contrast, MLR isolates target-instance consistency as a dedicated evaluation unit, making instance-consistency failures directly measurable rather than subsuming them into a broader composite score.

WFM-Eval. WFM-Eval explicitly diagnoses object hallucination and temporal inconsistency by comparing a generated video $\hat{V}$ with a ground-truth reference video V^* (Khose et al., 2026). It distinguishes phantom appearance, phantom disappearance, position drift, and frame jump. Let n_{pa} , n_{pd} , n_{dr} , and n_{jp} denote the numbers of these detected events over M generated frames and K tracked objects. Its Object Hallucination Rate (OHR) and Harm-weighted Severity Score (HSS) are

$$\text{OHR}_{\text{WFM}} = \frac{n_{\text{pa}} + n_{\text{pd}}}{M}, \quad \text{HSS}_{\text{WFM}} = \min\left(1, \frac{\sum_{e \in \mathcal{E}} w(e)}{MK}\right), \quad (9)$$

where $w(e)$ assigns event-dependent severity weights. These metrics provide fine-grained reference-grounded diagnostics. MLR instead anchors evaluation to the observed initial target state and therefore requires no ground-truth future, enabling target-instance consistency to be evaluated when only the conditioning observation and generated rollout are available.

T2V-CompBench. Generative numeracy in T2V-CompBench is prompt-anchored (Sun et al., 2025). GroundingDINO counts each prompted object class in every evaluated frame; a class receives score 1 if the detected quantity matches the quantity specified in the text prompt and 0 otherwise. The frame score averages over object classes, and the video score averages over frames. This rule can be written as

$$S_{\text{num}} = \frac{1}{T} \sum_{t=1}^T \frac{1}{J} \sum_{j=1}^J \mathbb{I}[\hat{n}_{t,j} = n_j^{\text{prompt}}], \quad (10)$$

where $\hat{n}_{t,j}$ is the detected count of object class j at frame t , n_j^{prompt} is its prompt-specified quantity, and J is the number of queried object classes. This metric therefore evaluates frame-wise agreement with prompt-specified quantities rather than target-instance consistency with the observed target population throughout an interaction. MLR instead derives the expected target count from the conditioning observation itself, allowing target-instance consistency to be evaluated even when the instruction does not explicitly specify a quantity.

RoboWM-Bench. RoboWM-Bench evaluates a different property: embodied executability (Jiang et al., 2026). Predicted human-hand or robotic videos are converted into executable actions through human-centric retargeting or a robot-centric inverse dynamics model (IDM), respectively, and the resulting actions are executed in simulation. Evaluation combines step-level checks at semantically meaningful action nodes with final task-level success. If C_r denotes whether key action node r satisfies its required interaction and dynamical constraints and G denotes achievement of the final task objective, its stated task-success criterion can be summarized as

$$S_{\text{task}} = \mathbb{I}\left[\left(\bigwedge_{r=1}^R C_r\right) \wedge G\right]. \quad (11)$$

This protocol measures whether generated behavior can be translated into successful embodied execution rather than directly measuring target-instance consistency in the generated rollout. MLR provides a complementary model-side diagnostic by directly identifying persistent target-count violations in the generated video, without requiring action decoding or simulator execution.

Positioning and Design Advantages of MLR. Taken together, these evaluations address complementary aspects of video and embodied-world-model quality but rely on different evaluation anchors. RBench embeds appearance and disappearance within broader physical and task-level assessment; WFM-Eval evaluates hallucination events relative to a ground-truth future; T2V-CompBench evaluates quantities explicitly specified by the prompt; and RoboWM-Bench assesses downstream embodied

execution. MLR is designed for the complementary regime in which the conditioning observation itself defines the expected target count and no reference future is assumed. It therefore provides four properties that are particularly useful for embodied manipulation evaluation: it is *initial-state anchored*, because the observed initial target count N_0 is obtained directly from the conditioning state; *reference-free*, because no ground-truth future is required; *rollout-level and persistence-aware*, because sustained violations are distinguished from isolated frame-level fluctuations; and *occlusion-aware*, because robot-supported under-counts are explicitly exempted. Together, these properties make MLR a dedicated diagnostic for persistent target-instance consistency failures in embodied world-model rollouts.

We operationalize MLR as follows. For DreamGenBench, eligibility is determined once from the fixed conditioning frame with GroundingDINO (Liu et al., 2024a) and the same target query, yielding the shared set $\mathcal{U}_{63}$ used for every model. GroundingDINO then counts target instances at T uniformly sampled future timestamps in each generated video. Spatially redundant detections are merged; items with no detectable target in the conditioning frame are excluded from the shared set. We denote the eligible set by

$$\mathcal{D}_+ = \{x \in \mathcal{D} : N_0(x) > 0\}.$$

We first identify deviations from the initial target-instance count N_0 . Over-counts, i.e., $N_t > N_0$, are always retained as duplication evidence. For under-counts, i.e., $N_t < N_0$, we further check whether the missing target is supported by robot-induced occlusion.

SAM2 (Ravi et al., 2025) tracks each target instance across the rollout. When the current target mask of an instance is unavailable, we use its most recent reliable mask as the reference target region. GroundingDINO localizes the robot in the current frame, and SAM2 converts the detected robot boxes into an occluder mask M_t^{robot} . For each missing target instance i , we compute the target–robot overlap ratio

$$r_{i,t}^{\text{occ}} = \frac{|\widehat{M}_{i,t}^{\text{target}} \cap M_t^{\text{robot}}|}{|\widehat{M}_{i,t}^{\text{target}}|}, \quad (12)$$

where $\widehat{M}_{i,t}^{\text{target}}$ denotes the reliable target reference region used for the occlusion check.

Motivated by prior amodal-segmentation criteria (Xiao et al., 2021; Zeng et al., 2022; Zhan et al., 2024) and target-normalized occlusion reasoning in robotic manipulation (Wada et al., 2018), we set $\tau_{\text{occ}} = 0.15$ and treat an under-count as occlusion-supported only if $r_{i,t}^{\text{occ}} \geq \tau_{\text{occ}}$ for every missing target instance; Appendix A.3 confirms that the comparison is robust to nearby thresholds. Such under-counts are excluded from violation evidence, whereas all other under-counts and all over-counts are retained. Let $o_t(x) = 1$ indicate an occlusion-supported under-count. We define the resulting evaluation count as

$$\tilde{N}_t(x) = \begin{cases} N_0(x), & o_t(x) = 1, \\ N_t(x), & \text{otherwise.} \end{cases} \quad (13)$$

For each eligible video, an MLR event is recorded when the adjusted count deviates from N_0 at two consecutive sampled timestamps:

$$e(x) = \mathbb{I}[\exists t \in \{1, \dots, T-1\} : \tilde{N}_t(x) \neq N_0(x) \wedge \tilde{N}_{t+1}(x) \neq N_0(x)],$$

$$\text{MLR} = \frac{100}{|\mathcal{D}_+|} \sum_{x \in \mathcal{D}_+} e(x). \quad (14)$$

Unlike frame-wise count agreement, MLR records a violation only when the inconsistency persists at the rollout level. An occlusion-supported under-count resets the persistence criterion, whereas a target that remains missing after the robot occluder moves away again provides disappearance evidence. Requiring two consecutive violating timestamps suppresses isolated detector errors while retaining sustained duplication and disappearance as consistency failures. All detector, query, occlusion-threshold, and persistence settings are fixed across models.

Beyond the scalar MLR value, the same event definition localizes each violation in time. For an MLR-positive rollout, we record the first persistent violation as its onset and analyze how long the

count deviation persists. These diagnostics distinguish short detector fluctuations from sustained consistency failures and enable direct inspection of the corresponding generated segment.

Algorithm 1 Occlusion-aware Model Laziness Rate evaluation. All detector, query, occlusion, and persistence settings are fixed across models.

Input: generated set $\mathcal{D}$, target-entity query $q(c)$, fixed detector G , sampled timestamp set $\mathcal{T}$.

Output: MLR, detector coverage, and per-video event labels.

1. For each video x , run G with $q(c)$ at all sampled timestamps, merge spatially redundant target detections, and obtain $N_t(x)$. If $N_0(x) = 0$, mark x as ineligible.
 2. If $N_t(x) > N_0(x)$, retain the deviation as duplication evidence. If $N_t(x) < N_0(x)$, proceed to the occlusion check.
 3. Track target instances with SAM2 and obtain the robot occluder mask in the current frame. For each missing instance, compute $r_{i,t}^{\text{occ}}$ using Equation 12.
 4. Exempt an under-count only if every missing target instance satisfies $r_{i,t}^{\text{occ}} \geq 0.15$. Set $\tilde{N}_t(x) = N_0(x)$ for such timestamps and $\tilde{N}_t(x) = N_t(x)$ otherwise.
 5. For every eligible video, set $e(x) = 1$ iff $\tilde{N}_t(x) \neq N_0(x)$ at two consecutive sampled timestamps; otherwise set $e(x) = 0$.
 6. Report $\text{MLR} = 100 \sum_{x \in \mathcal{D}_+} e(x) / |\mathcal{D}_+|$ and coverage $= 100 |\mathcal{D}_+| / |\mathcal{D}|$, together with event and eligible counts.
-



Figure 5: **Example of occlusion-aware MLR evaluation.** The rollout shows target detections and robot occlusion across sampled timestamps. Under-counts supported by sufficient target–robot overlap are excluded from violation evidence, whereas under-counts that persist without sufficient overlap are retained as target-disappearance evidence.

Occlusion example. Figure 5 illustrates how the occlusion check distinguishes temporary robot occlusion from target disappearance. During sufficient robot–target overlap, a missing target is excluded from violation evidence. Once the robot moves away, a target that remains missing is again treated as an under-count and contributes to an MLR event if the deviation persists across consecutive sampled timestamps.

A.2 SCOPE AND LIMITATIONS OF MLR

MLR is designed for manipulation scenarios that require instance consistency, including grasping, transport, placement, and rearrangement, where the observed initial target count N_0 is expected to remain unchanged throughout the interaction. The protocol does not assume $N_0 = 1$ and therefore applies to scenes containing multiple instances of the manipulated target.

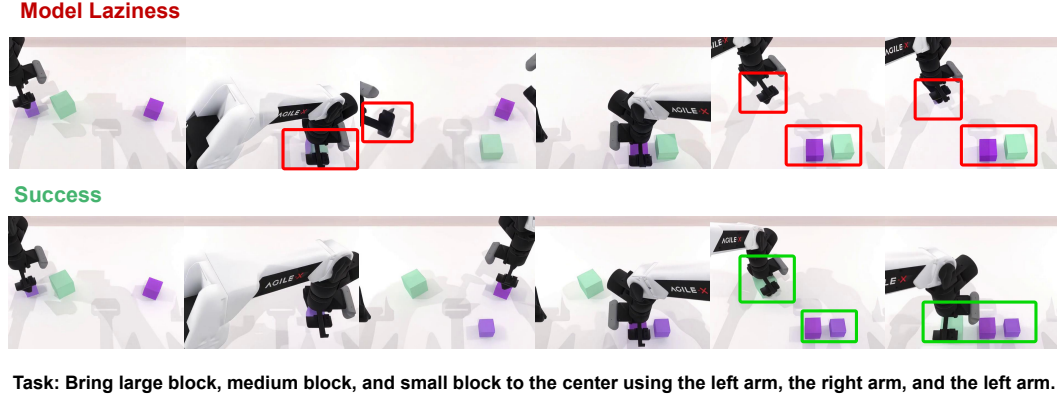
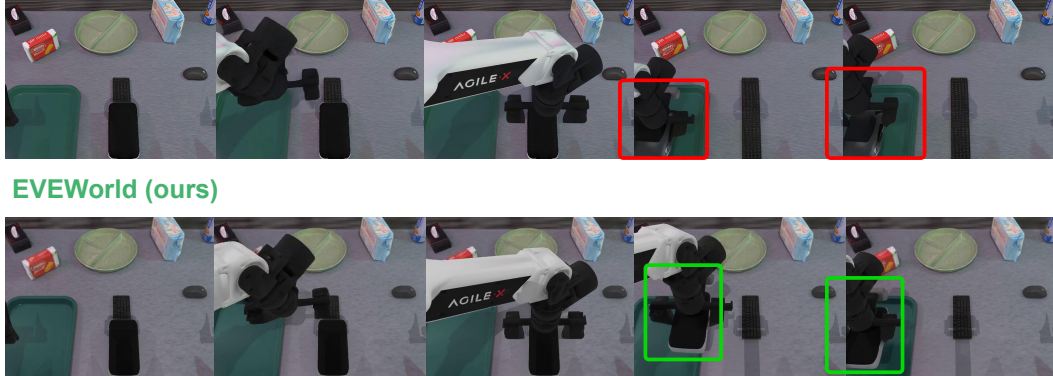


Figure 6: **Examples of MLR counting.** Red marks a persistent count violation retained by MLR. Green marks an under-count excluded by the occlusion-aware evaluation. A disappearance is counted only when under-count evidence persists outside occlusion-supported timestamps.

Interactions that intentionally change the target-instance count, such as cutting, splitting, assembly, or disassembly, fall outside the current formulation because the expected target count can change as part of successful task execution. Evaluating such tasks would require an instruction-conditioned expected count rather than a fixed initial-state reference.

MLR is also deliberately specific to target-instance consistency. Count-preserving failures, such as identity drift, deformation, or other changes that preserve the number of detected target instances, are not captured by MLR. These failures constitute a complementary form of temporal inconsistency and motivate the cross-frame supervision introduced by TIA.

Standard SFT | Model Laziness



Task: Place the rectangular black eraser for whiteboards on the tray with rounded corners, then set the mouse the rounded gray mouse next to the smooth black typing keyboard.

Figure 7: **Count-preserving cross-frame distortion.** Red marks Standard SFT, where the target deforms despite preserving its instance count. Green marks EVEWORLD, which preserves target shape and temporal consistency.

A.3 ROBUSTNESS AND SENSITIVITY ANALYSIS OF MLR

Because MLR is derived from automatic target counts, we examine its robustness to evaluation-time detection errors, robot-induced occlusion, and the temporal-persistence criterion. The final protocol combines spatial merging, occlusion-aware handling of under-counts, and temporal persistence, with all detector, query, and evaluation settings fixed across models.

Detector and occlusion robustness. Spatial merging suppresses redundant detections of the same target, while the occlusion check prevents robot-induced under-counts from being misclassified

as target disappearance. Only under-counts are eligible for the occlusion exemption; over-counts are always retained as duplication evidence. An occlusion-supported under-count is treated as non-violating and resets temporal persistence, whereas a target that remains missing after the robot occluder moves away again provides disappearance evidence.

Table 10 contrasts the raw count-deviation rate with the final MLR protocol. The substantial difference between raw detector deviations and the final event rate illustrates the importance of spatial merging, occlusion-aware filtering, and temporal persistence rather than treating every frame-level count mismatch as a physical-consistency failure.

Table 10: **Detector calibration for MLR.** Detector v1 denotes the raw count-deviation rate before spatial merging, occlusion-aware under-count handling, and temporal persistence. Peak dev. denotes the mean maximum raw count deviation, $\max_t |N_t - N_0|$, over eligible videos. MLR and Events / elig. are computed using the final occlusion-aware protocol.

Model / config.	Detector v1 (%)	MLR (%) ↓	Events / elig.	Peak dev.
GigaWorld-0 (Pretrained)	62.1	12.70	8 / 63	0.23
Official GR1 Fine-tune	68.8	7.94	5 / 63	0.14
Standard SFT	63.2	11.11	7 / 63	0.27
EVEWorld (step 100)	70.4	4.76	3 / 63	0.17
EVEWorld (seed 6666, step 250)	66.7	3.17	2 / 63	0.15
EVEWorld (u3, step 300)	68.0	3.17	2 / 63	0.03
EVEWorld (step 250, main)	70.4	1.59	1 / 63	0.11

Temporal persistence. For each eligible video, we record whether an adjusted count deviation $\tilde{N}_t(x) \neq N_0(x)$ persists for at least k consecutive sampled timestamps; $k=2$ is the default MLR rule. This persistence requirement suppresses isolated detector fluctuations while retaining sustained target duplication or disappearance.

Figure 8 reports the persistence length and cumulative onset of occlusion-aware target-instance consistency violations across models. The sole violation under EVEWORLD is confined to short persistence lengths, whereas violations in the general I2V baselines extend to larger k .

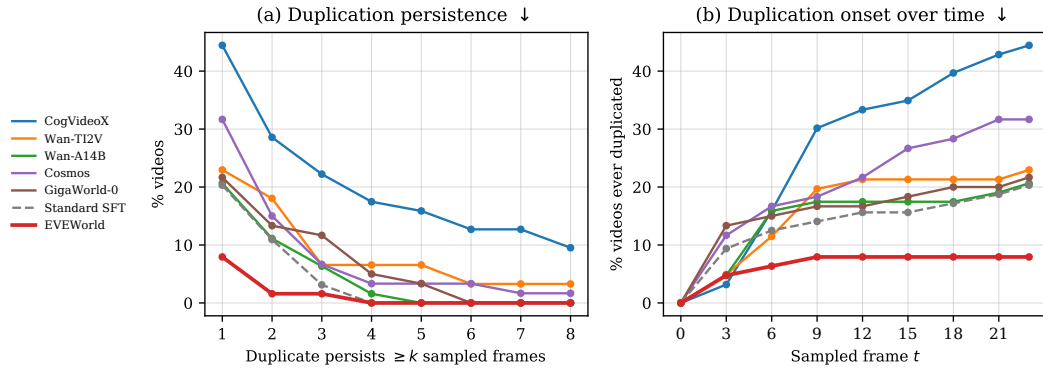


Figure 8: **Persistence of instance consistency violations across models.** (a) Fraction of detector-eligible DreamGenBench videos with occlusion-aware count violations lasting at least k consecutive sampled timestamps; (b) cumulative fraction of videos that have exhibited such a violation by each sampled timestamp.

Sensitivity to occlusion and persistence thresholds. The main protocol sets $\tau_{\text{occ}} = 0.15$, motivated by prior amodal-segmentation protocols that use approximately 15–17% occluded-area criteria to identify nontrivial occlusion (Xiao et al., 2021; Zeng et al., 2022; Zhan et al., 2024), and by target-normalized inter-instance occlusion reasoning in robotic manipulation (Wada et al., 2018).

We recompute MLR on the same shared set $\mathcal{U}_{63}$ with $\tau_{\text{occ}} \in \{0.10, 0.15, 0.20, 0.25\}$ and $k \in \{1, 2, 3\}$, while keeping all generated videos, detections, tracks, masks, queries, and eligible items

fixed. Across all configurations, EVEWORLD maintains a lower MLR than Standard SFT, with relative reductions ranging from 70.0% to 88.9%. Under the main setting, MLR decreases from 7/63 (11.11%) to 1/63 (1.59%), corresponding to an 85.7% relative reduction.

Lower τ_{occ} exempts more under-counts as robot-supported occlusions, whereas larger τ_{occ} retains more under-counts as MLR candidates; increasing k removes short-lived deviations. Neither change reverses the comparison between Standard SFT and EVEWORLD.

Table 11: **Sensitivity of MLR to the occlusion-overlap threshold τ_{occ} and persistence requirement k .** Each cell reports events / 63 (MLR in %). All settings use the same eligible set $\mathcal{U}_{63}$. The main setting, $\tau_{\text{occ}} = 0.15$ and $k = 2$, is shaded.

τ_{occ}	Standard SFT			EVEWorld		
	$k = 1$	$k = 2$	$k = 3$	$k = 1$	$k = 2$	$k = 3$
0.10	11 / 63 (17.46)	5 / 63 (7.94)	4 / 63 (6.35)	2 / 63 (3.17)	1 / 63 (1.59)	1 / 63 (1.59)
0.15	14 / 63 (22.22)	7 / 63 (11.11)	5 / 63 (7.94)	3 / 63 (4.76)	1 / 63 (1.59)	1 / 63 (1.59)
0.20	17 / 63 (26.98)	9 / 63 (14.29)	6 / 63 (9.52)	3 / 63 (4.76)	1 / 63 (1.59)	1 / 63 (1.59)
0.25	20 / 63 (31.75)	10 / 63 (15.87)	7 / 63 (11.11)	4 / 63 (6.35)	3 / 63 (4.76)	1 / 63 (1.59)

A.4 CROSS-MODEL EVALUATION OF MLR

We apply the same occlusion-aware MLR protocol across model families on DreamGenBench. For the controlled cross-model comparison, all event labels are evaluated on the same shared set of 63 detector-eligible items. Table 12 reports event counts, eligible counts, 95% Wilson confidence intervals, and coverage.

Across the evaluated model families, MLR ranges from 7.94–28.57%; CogVideoX1.5, Wan2.2-TI2V, Cosmos-Predict2, and pretrained GigaWorld-0 reach 28.57%, 17.46%, 14.29%, and 12.70%, respectively. These results show that persistent target-instance consistency failures occur across multiple video-generation and embodied-world-model families rather than being specific to Standard SFT or the GigaWorld backbone. On the same shared evaluation set, EVEWORLD attains an MLR of 1.59%.

Table 12: **Cross-model MLR on DreamGenBench.** Events / eligible denotes the number of Model Laziness cases over the shared 63-item set. 95% Wilson CI denotes the confidence interval of MLR, and Coverage denotes the fraction of detector-eligible prompts.

Model	Events / eligible	MLR (%) ↓	95% Wilson CI	Coverage (%)
GigaWorld-0 (Pretrained)	8 / 63	12.70	[6.58, 23.11]	50.00
Official GR1 Fine-tune	5 / 63	7.94	[3.44, 17.27]	50.00
Standard SFT	7 / 63	11.11	[5.49, 21.20]	50.00
Wan2.2-TI2V-5B	11 / 63	17.46	[10.04, 28.62]	50.00
Wan2.2-I2V-A14B	7 / 63	11.11	[5.49, 21.20]	50.00
CogVideoX1.5-5B-I2V	18 / 63	28.57	[18.90, 40.70]	50.00
Cosmos-Predict2-2B	9 / 63	14.29	[7.70, 24.97]	50.00
EVEWorld	1 / 63	1.59	[0.28, 8.46]	50.00

A.5 CROSS-DOMAIN EVALUATION OF MLR

We further evaluate the same occlusion-aware MLR protocol on WorldArena 1.0 (Shang et al., 2026a) using the officially released evaluation code locally. An instruction-only parser first selects manipulation prompts with an identifiable target (817 of 1,000), after which we apply the same initial-frame eligibility rule and MLR protocol used on DreamGenBench.

Table 13 reports the same cross-domain comparison as main-text Table 3, on the shared set of 157 detector-eligible videos and with 95% Wilson confidence intervals for every MLR value. MLR decreases from 27.39% under Standard SFT to 13.38% under EVEWORLD, showing that the

reduction in persistent target-instance consistency failures extends beyond the DreamGen domain. On this shared set the two baselines coincide at 27.39% while violating on different videos.

Table 13: **Cross-domain MLR** on WorldArena 1.0. Results use the same occlusion-aware protocol and the same shared set of 157 detector-eligible videos as main-text Table 3, with 95% Wilson confidence intervals.

Model	Events / eligible	MLR (%) ↓	95% Wilson CI	Coverage (%)
GigaWorld-0 (Pretrained)	43 / 157	27.39	[21.02, 34.84]	19.22
Standard SFT	43 / 157	27.39	[21.02, 34.84]	19.22
EVEWorld	21 / 157	13.38	[8.92, 19.58]	19.22

A.6 OPTIONAL VLM EVALUATION

For settings where a capable multimodal model is available, VLM-based temporal reasoning can provide an optional review of detector-based MLR events. This review is auxiliary: it neither replaces nor modifies the frozen detector-based MLR protocol, nor does it change the MLR labels or values reported in this paper. Because VLMs differ in visual and temporal reasoning capability, their judgments are treated as supporting evidence rather than ground truth, with manual review for ambiguous cases. Our qualitative distortion analysis uses the same framework with additional manual verification; examples are provided in Appendix F.3.

VLM Inputs and Outputs. When reviewing the same evaluation set, the eligible set $\mathcal{D}^+$, target query $q(c)$, and initial target count N_0 are inherited from the frozen primary MLR protocol. The VLM can be provided with the conditioning frame, instruction, target query, N_0 , frames aligned with the MLR sampling timeline, and a denser temporal view of the rollout. A JSON schema can constrain the output, including event type, interval, supporting frames, visibility explanation, and confidence.

MLR Judgment. For each candidate event, the VLM reviews the temporal evidence and returns the structured judgment above. Duplication and disappearance are mapped back to the fixed MLR sampling timeline and counted only when they satisfy the original persistence rule. Benign visibility loss, including supported robot or non-robot occlusion, temporary out-of-view motion, cropping, or other temporally supported visibility changes, is flagged as potential false positives of the primary detector-based MLR protocol and reported separately; they do not alter the primary MLR values. Distortion is excluded from MLR because the target count is preserved, but retained as a separate cross-frame consistency failure. Uncertain cases can be passed to manual review, with repeated judgments or a refutation query optionally used to check decision stability.

B METHOD DETAILS

B.1 IGR CONSTRUCTION

IGR constructs a corrupted–clean training pair from each valid target trajectory. An offline language parser extracts the target, source, and destination from the instruction. GroundingDINO (Liu et al., 2024a) localizes the target in sampled frames, and temporal association links valid detections into a trajectory. IGR then extracts an instance patch p and pastes one additional target instance (Ghiasi et al., 2021) at a location Ω that stays inside the frame, does not overlap the tracked target, and passes the occupancy and collision gates. If no valid paste region exists, we use the clean video as both input and target. With x the clean video, the corrupted input and the restoration target are

$$\tilde{x} = \text{Paste}(x, p, \Omega), \quad z = \text{VAE}(x), \quad \tilde{z} = \text{VAE}(\tilde{x}). \quad (15)$$

The model receives $\tilde{z}$ and reconstructs z . We use the final v4 restoration-region annotation shown in Figure 9 to define a binary spatial emphasis mask M_{rest} . The mask covers the target trajectory and task-relevant interaction regions around the source, destination, robot arm, and gripper, and additionally includes the actual pasted-instance region. Let

$$M(p) = \mathbb{I}[p \in \mathcal{R}_{\text{v4}} \cup \mathcal{R}_{\text{paste}}], \quad (16)$$

where p indexes a spatiotemporal latent location. Before normalization, we define

$$W(p) = 1 + 2M(p), \quad (17)$$

so that restoration-related locations receive three times the reconstruction weight of ordinary locations. We then normalize the map to unit mean,

$$\hat{W}(p) = \frac{W(p)}{\frac{1}{|\Omega_{\text{lat}}|} \sum_{q \in \Omega_{\text{lat}}} W(q)}, \quad (18)$$

where Ω_{lat} denotes all spatiotemporal locations in the latent grid. This normalization preserves the 3:1 relative weighting while keeping the global loss scale unchanged.

For DreamGen training, we propose copy-paste corruption with probability $p_{\text{dup}} = 0.5$. A proposal is applied only when the target trajectory and paste region pass the localization, occupancy, and collision gates; otherwise the sample falls back to the clean input. Consequently, the realized corrupted-sample fraction is lower than the proposal probability and is determined jointly by p_{dup} and the feasibility gates.

Algorithm 2 Offline construction of an IGR supervision pair. Invalid or unselected corruptions fall back to the clean training example.

Input: clean video x , instruction c , corruption probability p_{dup} , offline parser P , grounder G .

Output: corrupted input $\tilde{x}$, clean target x , and normalized regional weight map $\hat{W}$.

1. Parse (target, source, destination) $\leftarrow P(c)$; return the clean example if the parse is invalid.
 2. Ground the target on sampled frames with G and temporally associate valid detections into a trajectory $\mathcal{T}$; return the clean example if the confidence or continuity gate fails.
 3. Draw the corruption decision with probability p_{dup} ; return the clean example if corruption is not selected, otherwise extract a target patch from a reliable trajectory observation.
 4. Enumerate paste regions that remain in-frame, do not overlap the tracked target, and satisfy the occupancy and collision gates; return the clean example if this set is empty.
 5. Sample a valid region Ω , paste the additional instance into a copy of the video to obtain $\tilde{x}$, and retain the original video x as the restoration target.
 6. Assign weight 3 to the corrupted interaction region and weight 1 elsewhere, then normalize the resulting map to unit mean to obtain $\hat{W}$.
-

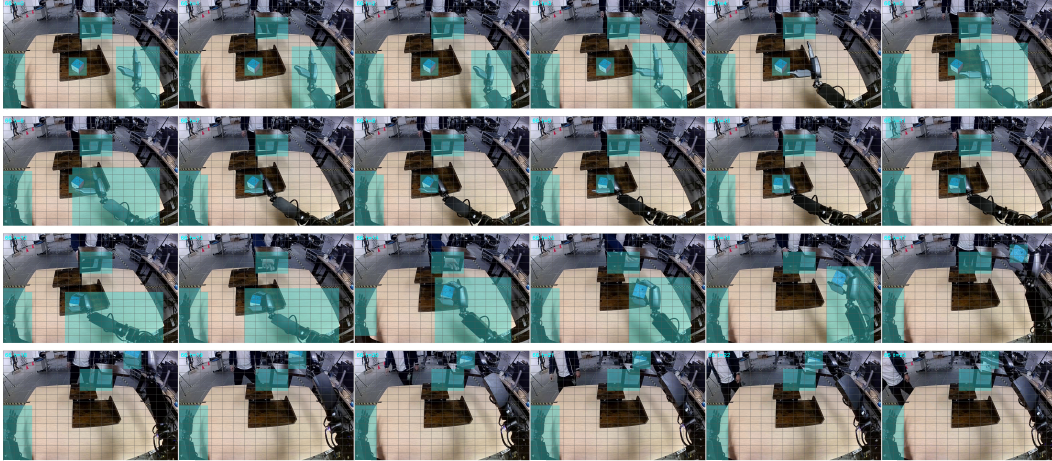


Figure 9: **Regional weighting in IGR.** The final v4 restoration region and the actual pasted-instance region receive three times the reconstruction weight of ordinary regions before unit-mean normalization.

Spatial-Weight Design Selection. We compared several spatial-emphasis designs before fixing the final IGR weighting. First, we tested a multi-level map that assigned different relative weights to the

Table 14: **Comparison of spatial-weight designs for IGR.** We vary both the spatial support of restoration supervision and its relative weighting strength. Legacy uses the original five-level weighting, while the remaining variants use binary weighting. Path+Loc restricts supervision to the target trajectory and source/destination regions; Interaction additionally covers robot-arm, gripper, and other manipulation-relevant regions. Coverage denotes the fraction of spatiotemporal latent locations assigned an emphasized restoration weight. MLR and IF are both reported in percentages, where lower MLR and higher IF are better. The Development columns compare candidate designs under a shared internal development protocol and are not directly comparable to the final results reported in the main text.

Design	Spatial Support			Relative Weight		Coverage	Development	
	Path/Loc.	Arm/Grip.	Paste	Base	Emph.	Median	MLR ↓	IF ↑
Legacy multi-level	✓	✓	✓	0.5	2/3/4/6	0.2679	2.18	60.31
Binary-3	✓	✓	✓	1	3	0.2685	2.21	60.26
Path+Loc-2x	✓		✓	1	2	0.2380	3.34	59.82
Path+Loc-3x	✓		✓	1	3	0.2147	3.08	59.96
Interaction-2x	✓	✓	✓	1	2	0.5361	2.06	60.42
Interaction-3x	✓	✓	✓	1	3	0.5361	1.94	60.57

target path, destination, vacated foreground, gripper, distractors, and pasted instance, using levels of 0.5/2/3/4/6. We also tested a simpler $3\times$ mask restricted to the target path and locations. The latter was more sensitive to incomplete localization because restoration errors are not confined to these two geometric regions: they can also occur near the gripper, along the robot-arm motion corridor, and around the source and destination regions.

We therefore refined a broader binary restoration mask while keeping a single emphasis factor. The v1 annotation covers the arm–object interaction region broadly; v2 removes irrelevant background-person regions and tightens the foreground support; and v4 corrects the remaining missed arm regions identified during annotation inspection. Table 14 reports the resulting coverage statistics. In our development comparisons, the multi-level weighting and the binary $3\times$ design yielded comparable average performance. We therefore adopt the simpler v4 binary weighting in the final model: ordinary locations receive relative weight 1, restoration-related locations receive relative weight 3, and the actual pasted-instance region uses the same relative weight 3. This design avoids excessively large local weights while retaining coverage of the regions in which duplicate-restoration errors can arise.

B.2 TIA ALIGNMENT

TIA consists of local cross-frame alignment and feature transport at a selected backbone layer ℓ^* . We first describe the alignment mechanism and then the offline consistency probe used to select ℓ^* across different domains and backbones. Throughout this section, $f_t^- = \text{norm}(W_{\text{in}}H_t^{\ell^*})$ denotes the projected and ℓ_2 -normalized feature used for locating and transport, and f_t^+ the ℓ_2 -normalized refined feature after transport, written $\bar{f}^{\ell^*}$ in the main text. The two stages use different windows: transport scores are computed over the dense window $\mathcal{C}_t^{\text{tr}}$ centered at the same coordinate for every cell, whereas the consistency loss is supervised over the window $\mathcal{C}_t$ centered at the true next-frame location u_t .

Cross-frame alignment and feature transport. At the selected layer ℓ^* , TIA establishes local target consistency between each pair of adjacent frames (Tang et al., 2023; Zhang et al., 2023; Oquab et al., 2024; Doersch et al., 2023; Karaev et al., 2024). A shared linear layer projects features from both frames into a normalized d -dimensional space. The feature at the previous target location u_{t-1} then queries the local window $\mathcal{C}_t^{\text{tr}}$ centered at the same coordinate in frame t , which yields the pre-transport similarity:

$$s_{t,v}^- = f_{t-1}^-(u_{t-1})^\top f_t^-(v). \quad (19)$$

For feature transport, we apply the local matcher densely. Each cell in frame $t-1$ queries a local window centered at the same coordinate in frame t ; a softmax with temperature τ converts the similarities into transport weights (van den Oord et al., 2018). The projected frame- $(t-1)$ features are

splatted into frame t using these weights, normalized per cell by the accumulated mass, and written back through a bounded residual update:

$$H'_t = H_t^{\ell^*} + \gamma \tanh(W_{\text{out}}(\bar{H}_t - W_{\text{in}}H_t^{\ell^*})), \quad (20)$$

where $\bar{H}_t$ is the transported feature field, γ controls the residual magnitude, and W_{out} is a bias-free linear layer initialized to zero. The first frame is left unchanged. The consistency loss is computed on the refined features after this update: we re-normalize H'_t to obtain $f_t^+ = \text{norm}(H'_t)$, recompute the similarities $s_{t,v}^+ = f_{t-1}^+(u_{t-1})^\top f_t^+(v)$ over the window $\mathcal{C}_t$ centered at the true next-frame location u_t , treat u_t as the positive target and the remaining candidates as negatives, and optimize these correspondences with the temperature-scaled consistency loss $\mathcal{L}_{\text{TIA}}$ defined in the main text. The transport weights and the consistency loss share the temperature τ but are computed in separate passes.

Algorithm 3 TIA training-time consistency and feature transport.

Input: adjacent-frame hidden features $(H_{t-1}^{\ell^*}, H_t^{\ell^*})$, target locations (u_{t-1}, u_t) , shared down-projection W_{in} , zero-initialized output projection W_{out} , transport window $\mathcal{C}_t^{\text{tr}}$, supervision window $\mathcal{C}_t$ centered at u_t , temperature τ , residual scale γ .

Output: updated current-frame features H'_t , consistency loss $\mathcal{L}_{\text{TIA}}$.

1. Project both frames with W_{in} and normalize the resulting features to obtain f^- . Each cell in frame $t-1$ forms a query, and the local window around the corresponding coordinate in frame t forms its candidate set $\mathcal{C}_t^{\text{tr}}$.
 2. Compute the pre-transport similarities $s_{t,v}^-$ within each local window and convert them into transport weights using a softmax with temperature τ .
 3. Splat the projected frame- $(t-1)$ features into frame t according to these weights, normalize each destination cell by its accumulated mass, and apply the residual update in Equation (20). The first frame is left unchanged.
 4. Normalize the updated features H'_t to obtain f^+ , recompute the similarities $s_{t,v}^+$ within the supervision window $\mathcal{C}_t$, treat u_t as the positive target and the remaining candidates as negatives, and compute the temperature-scaled consistency loss $\mathcal{L}_{\text{TIA}}$.
 5. Back-propagate $\mathcal{L}_{\text{TIA}}$ through the TIA sidecar and the trainable backbone.
-

Layer selection via consistency probing. We select the TIA attachment layer using an offline consistency probe rather than fixing it heuristically. For each candidate layer, the target feature in the previous frame retrieves its nearest match over the full token grid of the next frame, using the raw ℓ_2 -normalized block features rather than the W_{in} -projected features used inside TIA, and endpoint error (EPE) (Teed & Deng, 2020) is measured against the localized target position. Because feature reliability can differ across domains and backbone architectures, we additionally report a setting-specific consistency-reliability measure. Table 15 summarizes the probe results and the resulting layer choices for GigaWorld-0/DreamGen, AgiBot, and FlowWAM.

Table 15: **Layer-wise consistency probe** across domains and backbones. EPE denotes the mean endpoint error of adjacent-frame target matching in token cells. Reliab. is the setting-specific consistency-reliability score: object-background separation for DreamGen, static-feature stability for AgiBot, or moving-pair top-1 hit rate for FlowWAM. Block 23 is selected for GigaWorld-0/DreamGen and AgiBot, and block 12 is selected for FlowWAM.

Block	GigaWorld-0 / DreamGen		AgiBot		FlowWAM	
	EPE ↓	Reliab. ↑	EPE ↓	Reliab. ↑	EPE ↓	Reliab. ↑
8	0.80	−0.09	9.24	0.72	3.50	0.172
9	0.80	−0.07	8.88	0.74	3.56	0.171
10	0.93	−0.09	8.00	0.73	3.46	0.178
11	0.93	−0.09	7.52	0.74	3.45	0.176
12	0.98	−0.08	6.82	0.74	3.42	0.182
13	0.68	−0.07	6.56	0.75	3.46	0.177
14	0.98	−0.05	6.48	0.78	3.51	0.174
15	0.98	−0.10	6.39	0.81	3.49	0.172
16	0.73	−0.06	6.44	0.81	3.59	0.165
17	0.78	−0.06	6.62	0.85	3.64	0.158
18	0.98	−0.07	6.54	0.83	3.51	0.158
19	0.91	−0.07	6.64	0.83	3.56	0.155
20	0.78	−0.07	6.22	0.83	3.60	0.154
21	0.75	−0.10	6.44	0.80	3.65	0.146
22	0.63	−0.16	6.98	0.83	3.69	0.135
23	0.55	−0.10	5.72	0.76	3.68	0.144
24	1.08	−0.19	9.81	0.81	3.71	0.139
25	0.96	−0.20	7.53	0.71	3.89	0.126
26	1.28	−0.24	9.79	0.65	3.87	0.124

B.3 JOINT TRAINING AND INFERENCE

The full training objective combines the IGR restoration loss with the TIA consistency loss:

$$\mathcal{L} = \mathcal{L}_{\text{IGR}} + \lambda_{\text{TIA}} \mathcal{L}_{\text{TIA}}. \quad (21)$$

For samples with an applied IGR corruption, $\mathcal{L}_{\text{IGR}}$ restores the clean latent from the count-perturbed input; otherwise, it reduces to the base clean-video training objective. The consistency term is computed on the post-transport features. Its weight and optional noise gate are set per backbone in Appendix C.1.

At inference, the model receives the first frame, the instruction, and the benchmark-specific control input; the parser, the grounder, and the corruption generator are training-time only.

C ADDITIONAL EXPERIMENTS AND ANALYSIS

C.1 IMPLEMENTATION DETAILS

Within each evaluation arm, EVEWORLD and its control use the same initialization, training set, and optimization budget. The DreamGenBench and EWMBench experiments initialize from GigaWorld-0 Video-Pretrain-2B and update the full transformer with CAME-8bit (Luo et al., 2023), a learning rate of 4.32×10^{-5} , and an effective batch size of 64. For DreamGenBench, we train on 92 GR1 videos for 250 steps at 480×768 resolution using 93-frame clips at 16 FPS, and evaluate the raw step-250 checkpoint. For EWMBench, we train on 777 AgiBot clips for 50 steps at 480×640 resolution with the same frame count and frame rate, and evaluate the raw step-50 checkpoint.

For RoboTwin, both models initialize from the released FlowWAM Stage-1 checkpoint and train a rank-32 LoRA (Hu et al., 2022) on 2,250 episodes for 1,128 optimization steps. We use AdamW (Loshchilov & Hutter, 2019) with a learning rate of 10^{-4} , weight decay 0.01, and batch size 8. Training clips contain 29 frames at 24 FPS with a resolution of 480×640 ; evaluation uses the final-epoch checkpoint.

In the evaluated instantiations, TIA uses a 64-dimensional projection, 7×7 local windows, temperature $\tau = 0.07$, and residual scale $\gamma = 0.1$. It attaches to block 23 on GigaWorld-0 and AgiBot, and block 12 on FlowWAM. On GigaWorld-0, λ_{TIA} warms from 0 to 0.5 over the first 20 steps and applies at $\sigma \in [0.2, 0.5]$; FlowWAM uses $\lambda_{\text{TIA}} = 0.1$ without noise gating.

DreamGenBench and WorldArena use the raw step-250 checkpoint, 30 denoising steps, and CFG 7.0 at 480×768 resolution with 93 frames at 16 FPS. EWMBench uses the raw step-50 checkpoint, the three benchmark seeds, 30 steps, and CFG 7.0 at 480×640 with the same clip length and frame rate. RoboTwin uses the final-epoch checkpoint and CFG 5.0 at 480×640 , generating the full trajectory at 24 FPS. All settings condition on the first frame and instruction; RoboTwin additionally uses robot-only optical flow. Appendix C.4 studies CFG and denoising choices separately.

C.2 IGR ABLATION AND DUPLICATE RESTORATION

Restoration-objective ablation. Table 16 isolates the effects of duplicate-restoration supervision and spatial weighting. Removing spatial emphasis from the restoration objective still reduces MLR relative to Standard SFT, but leaves substantially more violations than the complete EVEWORLD. Conversely, applying spatial emphasis without the duplicate-restoration target does not improve instance consistency. IGR alone reduces MLR to 4.76%, while the complete EVEWORLD further reduces it to 1.59%. Notably, removing spatial emphasis yields a slightly higher Overall instruction-following score than the complete model (61.11% vs. 60.85%) but increases MLR from 1.59% to 6.35%, supporting the use of the complete objective.

Table 16: **IGR restoration controls** on DreamGenBench. The first two variants isolate spatial weighting and duplicate-restoration supervision. IGR w/o spatial emphasis removes the spatial weighting, and Spatial emphasis only applies the weighting without the duplicate-restoration target. Env, Object, Behavior, and Overall report Gemini-IF instruction-following accuracy in percentages.

Variant	MLR (%) ↓	Env ↑	Object ↑	Behavior ↑	Overall ↑
IGR w/o spatial emphasis	6.35	56.32	44.00	82.27	61.11
Spatial emphasis only	14.29	34.48	34.67	62.41	44.97
IGR only	4.76	49.43	36.67	69.50	51.85
EVEWORLD	1.59	72.41	50.33	64.89	60.85

Controlled duplicate-restoration evaluation. We further test whether IGR learns to actively correct a corrupted target count rather than merely preserving the input appearance. We construct an independent corruption set by inserting one additional target instance into clean demonstrations and reconstruct the resulting videos under six combinations of paste opacity α and diffusion noise level σ . Starting from the Standard SFT checkpoint, the model preserves 92–99% of the injected area, indicating that the spurious instance is largely retained. After 50 IGR fine-tuning steps, retention decreases to 23–42% across all six settings (Table 17). The directional cosine decreases together with retention, showing that the residual reconstruction error becomes not only smaller but also less aligned with the injected duplicate. Figure 10 summarizes both effects across corruption strengths.

Table 17: **Robustness of duplicate removal** across corruption strengths. Retention is the fraction of injected duplicate area preserved by the reconstruction. Dir. cos. measures the alignment between the residual error and the injected duplicate. Before and after denote the Standard SFT checkpoint and the model after 50 IGR fine-tuning steps, respectively.

Opacity α	σ	Ret. before ↓	Ret. after ↓	Rel. reduction (%) ↑	Dir. cos. before / after
0.5	0.3	0.992	0.421	57.5	0.983 / 0.627
0.5	0.8	0.955	0.325	66.0	0.950 / 0.378
0.5	1.5	0.922	0.275	70.2	0.907 / 0.263
1.0	0.3	0.986	0.419	57.5	0.993 / 0.695
1.0	0.8	0.968	0.268	72.3	0.980 / 0.406
1.0	1.5	0.935	0.229	75.5	0.961 / 0.295

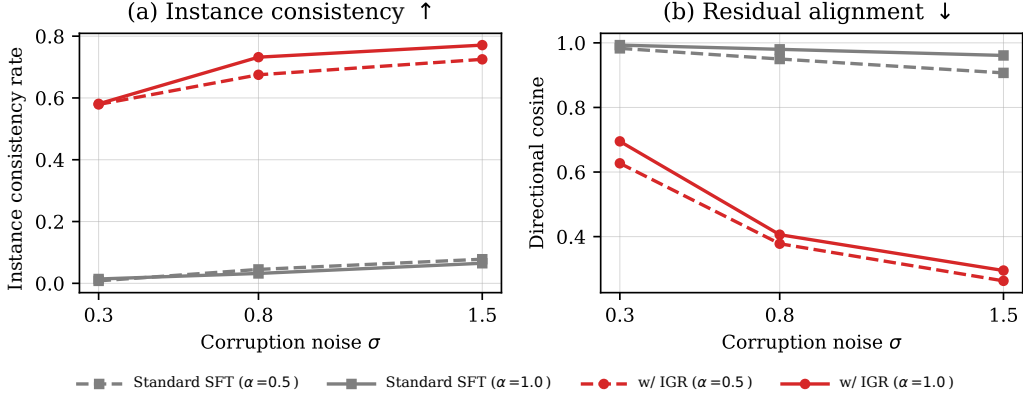


Figure 10: **Controlled restoration analysis of IGR.** Injected duplicates probe restoration behavior under different corruption-noise levels σ and paste opacities α . (a) Standard SFT largely retains the injected duplicate, whereas IGR substantially improves duplicate removal across corruption strengths. (b) Residual errors under Standard SFT remain strongly aligned with the injected duplicate, while IGR reduces this alignment after restoration training.

Feature diagnostics. Two additional diagnostics use the same feature-probe family as Appendix B.2. Blocks 25–26 best separate changed from unchanged regions, with score gaps of 0.43–0.45, while blocks 10–16 provide the strongest detection of injected duplicates, with AUC values of 0.79–0.80.

C.3 EVALUATION STABILITY

We quantify variability from repeated instruction-following judgments and inference seeds. Table 18 reruns Gemini-IF on fixed videos and reports every repeat underlying each mean; the main ordering remains unchanged. Table 19 further compares Standard SFT and EVEWORLD under matched inference seeds, explicitly controlling for sampling stochasticity. The consistent advantage of EVEWORLD indicates that the observed gains are not attributable to favorable seed selection.

Table 18: **Gemini-IF judge variance** across models on DreamGenBench. Overall reports mean $\pm$ SD over repeated Gemini-IF evaluations of the same generated videos. Env, Object, and Behavior report the corresponding mean scores on each split. Repeats denotes the number of independent Gemini-IF evaluations.

Model	Repeats	Gemini-IF (%) $\uparrow$			
		Overall	Env	Object	Behavior
CogVideoX1.5-5B-I2V (Yang et al., 2025)	2	5.56 $\pm$ 1.12	15.52	1.00	4.26
Wan2.2-TI2V-5B (Team Wan et al., 2025)	2	10.32 $\pm$ 1.12	17.24	5.00	11.70
Wan2.2-I2V-A14B (Team Wan et al., 2025)	2	15.87 $\pm$ 3.37	20.69	13.00	15.96
Cosmos-Predict2-2B (NVIDIA et al., 2025)	2	24.60 $\pm$ 1.12	31.03	28.00	17.02
GigaWorld-0 (GigaWorld Team et al., 2025)	6	60.19 $\pm$ 2.90	54.60	54.33	69.86
Official GR1 Fine-tune (Jang et al., 2025)	2	63.10 $\pm$ 6.17	55.17	46.00	86.17
Standard SFT	6	53.57 $\pm$ 2.92	51.72	42.00	67.02
EVEWorld	6	60.85 $\pm$ 1.79	72.41	50.33	64.89

Table 19: **Matched inference-seed comparison** of Standard SFT and EVEWORLD on DreamGenBench. Gemini-IF reports mean $\pm$ SD over repeated judge evaluations at each inference seed. Δ denotes the improvement of EVEWORLD over Standard SFT in percentage points.

Seed	Judge repeats	Standard SFT (%)	EVEWORLD (%)	Δ (pp)
004	6	53.57 $\pm$ 2.92	60.85 $\pm$ 1.79	+7.28
061	3	57.41 $\pm$ 2.79	58.73 $\pm$ 1.37	+1.32

Under the transfer protocol, five complete FlowWAM generations from the same checkpoint differ by at most 0.68 Local-11 points (Table 20), indicating stable overall performance across inference seeds. We further examine per-component variation across five inference seeds on a 50-prompt probe set (Table 21). Motion-related metrics show larger variation than appearance-related metrics, but the aggregate Local-11 score remains relatively stable. Local-11 denotes the project-local 11-metric evaluator.

Table 20: **Inference-seed variance** under the full-1000 transfer protocol. Local-11 is evaluated from the same FlowWAM checkpoint across five inference seeds. Mean $\pm$ SD reports variation across seeds.

Seed	6	12	20	42	70	Mean $\pm$ SD
Local-11	57.12	56.89	57.57	57.37	57.46	57.28 $\pm$ 0.28

Table 21: **Per-component inference-seed variance** under the transfer protocol. Component scores are evaluated across five inference seeds on a 50-prompt probe set. Mean $\pm$ SD reports variation across seeds.

Component	Seed 0	Seed 17	Seed 35	Seed 42	Seed 70	Mean $\pm$ SD
Image Quality	58.95	59.41	59.35	59.37	57.46	58.91 $\pm$ 0.83
Aesthetic	44.30	44.18	44.56	44.38	44.35	44.35 $\pm$ 0.14
Dynamic Degree	29.59	31.44	29.79	27.88	33.17	30.37 $\pm$ 2.01
Flow Score	18.65	20.57	17.97	15.67	20.11	18.59 $\pm$ 1.95
Motion Smoothness	68.64	68.79	68.14	68.15	71.28	69.00 $\pm$ 1.31
Subject Consistency	73.84	76.67	73.22	80.08	75.56	75.87 $\pm$ 2.72
Background Consistency	80.25	82.59	79.25	86.17	82.36	82.12 $\pm$ 2.67
Photometric	24.71	28.02	24.61	30.75	24.45	26.51 $\pm$ 2.80
Interaction	54.40	56.00	57.60	58.40	58.80	57.04 $\pm$ 1.82
Perspectivity	87.60	86.00	89.20	88.00	85.20	87.20 $\pm$ 1.60
Instruction Following	47.20	47.20	50.80	53.20	54.80	50.64 $\pm$ 3.45
Local-11	53.47	54.63	54.05	55.64	55.23	54.60 $\pm$ 0.88

C.4 TRAINING AND LONG-HORIZON BEHAVIOR

Training dynamics. We first examine how instance consistency and instruction following evolve during post-training. As training proceeds toward the selected step-250 checkpoint, MLR generally decreases while Qwen-IF improves, indicating that the reduction in Model Laziness is accompanied by better task fidelity rather than obtained at its expense.

Guidance sensitivity. We further evaluate classifier-free guidance (CFG) (Ho & Salimans, 2022) across multiple post-training checkpoints and guidance weights. Table 22 shows that stronger guidance generally raises Gemini-IF, but MLR is substantially less monotonic. For example, increasing w can improve instruction following while leaving MLR unchanged or even increasing it at the same checkpoint. Conversely, the lowest MLR values occur only for particular combinations of training stage and guidance weight rather than systematically at the strongest guidance setting. These results show that CFG primarily controls inference-time instruction fidelity, whereas target-instance consistency remains dependent on the learned post-training behavior.

Long-horizon behavior. Finally, we examine whether longer rollouts introduce broader physical and process-level degradation. On PBench (NVIDIA, 2025b), overall physical-QA performance decreases as the output duration grows from 3.8 s to 19.8 s, with temporal reasoning degrading most strongly (Table 23). Averaged over all five durations, Standard SFT improves all four PBench dimensions over the pretrained backbone. The advantage becomes more pronounced at longer horizons: over durations of at least 9.8 s, it improves Domain, Phys, and Time by 1.65, 1.95, and 4.24 points on average, respectively. These results indicate that post-training improves broader physical reasoning even as long-horizon generation remains increasingly challenging.

Table 22: **CFG sensitivity** across training stages on DreamGenBench. Each cell reports Gemini-IF $\uparrow$ / MLR $\downarrow$ in percentages under different CFG weights w and training steps. $w=1.0$ denotes inference without classifier-free guidance. The shaded column marks the step-250 checkpoint used in the main experiments. The shaded entry at step 250 and $w=7.0$ corresponds to the main EVEWorld configuration reported in main-text Table 1.

CFG weight w	Train step					
	50	100	150	200	250	300
1.0	26.08 / 9.52	39.32 / 6.35	26.91 / 4.76	26.08 / 9.52	28.98 / 9.52	31.87 / 6.35
2.5	45.95 / 4.76	54.23 / 11.11	61.68 / 4.76	47.19 / 3.17	50.91 / 4.76	55.06 / 4.76
5.0	50.91 / 7.94	57.53 / 9.52	57.53 / 3.17	51.32 / 1.59	58.78 / 6.35	48.85 / 4.76
7.0	52.15 / 4.76	66.23 / 9.52	60.43 / 6.35	53.40 / 1.59	60.85 / 1.59	45.95 / 3.17

Table 23: **PBench physical-QA accuracy** across output durations. Domain reports the overall PBench Robot score, judged by Qwen3.6-VL. Phys, Space, and Time report accuracy on physical, spatial, and temporal reasoning. Mean averages the five durations, and Mean (≥ 9.8 s) averages the three longest horizons.

Dur. (s)	GigaWorld-0 (Pretrained)				Standard SFT			
	Domain	Phys.	Space	Time	Domain	Phys.	Space	Time
3.8	82.43	85.71	84.24	82.18	80.98	86.69	83.33	79.64
5.8	79.87	84.42	82.12	74.18	79.89	85.71	83.94	74.91
9.8	74.32	83.77	76.06	65.09	78.90	87.99	79.70	74.55
15.8	67.54	79.22	73.03	49.82	68.12	79.55	70.61	53.82
19.8	66.97	76.95	74.55	50.18	66.76	78.25	73.94	49.45
Mean	74.23	82.01	78.00	64.29	74.93	83.64	78.30	66.47
Mean (≥ 9.8 s)	69.61	79.98	74.55	55.03	71.26	81.93	74.75	59.27

Longer horizons also expose failures that are not captured by target-instance count alone. On 92 matched prompts, both GigaWorld-0 variants exhibit substantially higher shortcut severity, shortcut frequency, and missing-stage frequency at 7.8 s than at 3.8 s (Table 24). At the shared 3.8 s horizon, Wan2.2-TI2V-5B exhibits still larger process errors than either GigaWorld-0 variant. These diagnostics complement MLR by measuring broader failures of process completeness rather than instance consistency alone.

Table 24: **Long-horizon process evaluation** across models and output durations. Qwen-based evaluation is performed on 92 matched prompts. Severity measures shortcut severity. Any shortcut and Missing stage report the fractions of generations containing a shortcut or a missing stage.

Model / horizon	Severity $\downarrow$	Any shortcut $\downarrow$	Missing stage $\downarrow$	n
GigaWorld-0 SFT, 3.8s	0.94	0.32	0.30	92
GigaWorld-0 SFT, 7.8s	1.79	0.63	0.60	92
GigaWorld-0 Pretrained, 3.8s	1.17	0.39	0.38	92
GigaWorld-0 Pretrained, 7.8s	1.97	0.67	0.66	92
Wan2.2-TI2V-5B, 3.8s	2.30	0.82	0.72	92

D EXPLORATION OF ALTERNATIVE APPROACHES

We evaluate five alternative designs spanning training to inference: PhysicsLatent, EAG, LAD-LoRA, Causal Frontier, and ICH-D. Because their training budgets and generation protocols differ, each route is compared with its own control using the same prompts, conditioning frames, and inference seeds. Table 25 reports MLR and Gemini-IF, with 95% confidence intervals computed by paired bootstrap over the matched pairs.

No alternative jointly reduces MLR and improves instruction following. PhysicsLatent slightly reduces MLR but lowers Gemini-IF, EAG improves Gemini-IF but increases MLR, and the remaining routes show no joint gain.

Table 25: **Comparison with alternative designs.** Each route is evaluated against its matched control using the same prompts, conditioning frames, and inference seeds. Panel (a) reports MLR for each route and its matched control. Panel (b) reports Gemini-IF as mean $\pm$ SD over three judge repeats. Rel. Δ denotes the relative change from the matched control, and 95% CIs are computed by paired bootstrap. MLR values are reported to one decimal place; Rel. Δ and the CIs are computed from unrounded values.

(a) Model Laziness Rate (MLR)					
Route	Matched control	MLR (%) $\downarrow$		Rel. Δ	95% CI
		Control	Route		
PhysicsLatent	Pretrain 5.8s	17.4	16.2	-7.0	$[-51.5, +69.1]$
EAG	weight 0	15.4	38.5	+150.0	$[0.0, +500.0]$
LAD-LoRA	LoRA control	3.8	3.8	0.0	$[0.0, 0.0]$
Causal Frontier	control50	30.8	41.7	+35.4	$[-42.9, +300.0]$
ICH-D	Pretrain (4 seeds)	11.2	13.1	+16.6	$[-19.4, +70.8]$

(b) Gemini-IF				
Route	Control	Route	Rel. Δ	95% CI
PhysicsLatent	45.65 ± 3.92	40.22 ± 3.26	-11.9	$[-29.8, +7.9]$
EAG	41.67 ± 3.61	45.83 ± 9.55	+10.0	$[-15.8, +54.5]$
LAD-LoRA	51.04 ± 1.80	47.92 ± 4.77	-6.1	$[-24.1, +13.3]$
Causal Frontier	27.08 ± 3.61	12.50 ± 6.25	-53.8	$[-100.0, +85.7]$
ICH-D	45.83 ± 1.55	31.79 ± 0.47	-30.6	$[-40.1, -20.2]$

D.1 PHYSICSLATENT

PhysicsLatent augments the conditioning sequence with eight learned process tokens. The tokens are computed from the pooled reference-frame latent and prompt embedding and are read by every cross-attention layer. Weak-label heads predict the target’s start and goal boxes, contact events, a coarse 2D trajectory, and the manipulation phase, using pseudo-labels parsed from the instruction together with color and motion heuristics. Against the pretrained 5.8 s control (92 matched pairs), MLR moves from 17.4% to 16.2% while Gemini-IF drops from 45.65 to 40.22, leaving no joint gain.

Algorithm 4 PhysicsLatent training-sample construction. Pseudo-labels come from instruction parsing and color/motion heuristics; low-confidence labels are masked out.

Input: training video x , instruction c , frozen VAE and text encoder, process-token encoder E_ψ , weak-label heads $\{h_k\}$.

Output: process-token-augmented input and auxiliary training targets.

1. Encode the reference frame with the VAE and pool it to a 15×24 grid; embed the instruction with the text encoder.
 2. Produce $K=8$ process tokens (1024-dimensional) from both with E_ψ and append them after the text tokens; every cross-attention layer reads them, and the two CFG branches share the same tokens.
 3. Predict the weak labels from the process tokens: start and goal boxes, contact events, an 8-keyframe 2D trajectory, the manipulation phase, and completion flags.
 4. Obtain pseudo-labels by parsing the instruction (target, source, destination) and applying color- and motion-based heuristics on the frames; mask out low-confidence labels.
 5. Train with $\mathcal{L}_{\text{video}} + \sum_k w_k \mathcal{L}_k$, where each w_k is small (0.03–0.08) so the auxiliary terms do not dominate reconstruction.
-

D.2 EAG

EAG modifies sampling without retraining the video model. After each CFG update, a frozen latent-action model (Schmidt & Jiang, 2024; Ye et al., 2025) assigns a transition energy $E(\hat{z}_0)$ to the predicted clean latent $\hat{z}_0$ using a soft top-3 aggregation of per-transition reconstruction errors. The sampler then takes one normalized step along the negative energy gradient:

$$\hat{z}_0 \leftarrow \hat{z}_0 - w(\sigma) \|\hat{z}_0\| \frac{\nabla_{\hat{z}_0} E(\hat{z}_0)}{\|\nabla_{\hat{z}_0} E(\hat{z}_0)\|}, \quad w(\sigma) = \frac{w_0}{1 + \sigma}, \quad (22)$$

where w_0 is the guidance weight and $w_0=0$ recovers the unmodified sampler. Against the weight-0 control, Gemini-IF rises from 41.67 to 45.83, while MLR rises from 15.4% to 38.5% (16 prompt-seed pairs, 13 detector-eligible). This small comparison is reported as a development result.

Algorithm 5 EAG-guided denoising step. Only the sampler changes; the model and the energy model stay frozen.

Input: denoiser D_θ , frozen transition-energy model E , scheduler, guidance weight w_0 .

Output: the guided denoising update at the current step.

1. Run the usual CFG update and form the clean-latent prediction $\hat{z}_0$ from the model output.
 2. Score $\hat{z}_0$ with the frozen transition energy $E(\hat{z}_0)$.
 3. Take one normalized step down the energy (Eq. (22)); the step length scales with $\|\hat{z}_0\|$, so the update is bounded.
 4. Convert the change back into the model output space, $\Delta_{\text{noise}} = \Delta\hat{z}_0/c_{\text{out}}(\sigma)$, and let the scheduler proceed as usual.
-

D.3 LAD-LoRA

LAD-LoRA moves the transition-energy signal from sampling to parameter updates. We train a rank-64 attention LoRA (Hu et al., 2022), with the backbone frozen, by adding the energy of the model’s denoised output as a regularizer:

$$\mathcal{L} = \mathcal{L}_{\text{EDM}} + \frac{\tilde{w}}{1 + \sigma} E(\hat{z}_{0,\theta}), \quad (23)$$

where $\tilde{w}$ is recomputed at every step so that the energy gradient stays at a fixed fraction of the denoising gradient on the shared prediction $\hat{z}_0$ (target ratio 0.1, capped at 1.0). Against a plain LoRA control under the same protocol (32 matched pairs), LAD-LoRA has the same MLR (3.8%) and reduces Gemini-IF from 51.04 to 47.92.

Algorithm 6 LAD-LoRA training step. The backbone is frozen; only the rank-64 attention LoRA receives gradients.

Input: frozen backbone with LoRA adapters, frozen transition-energy model E , target gradient ratio $w_{\text{lad}}=0.1$.

Output: LoRA parameter update.

1. Denoise the batch and form the model’s own clean prediction $\hat{z}_{0,\theta}$.
 2. Score $\hat{z}_{0,\theta}$ with the frozen energy E (the same soft top-3 transition energy as EAG).
 3. Measure the gradient norms of the denoising loss and of E on the shared $\hat{z}_0$, set the energy weight to hit the target gradient ratio, and gate by $1/(1 + \sigma)$.
 4. Back-propagate Eq. (23) into the LoRA parameters only.
-

D.4 CAUSAL FRONTIER

Causal Frontier generates a video autoregressively in blocks of four latent frames. At each step, the model sees only the clean latents of previously committed blocks, which a near-zero noise level and a mask channel mark as history. The denoising loss applies only to the active block. Generation commits each block irreversibly before denoising the next. At the current budget, prediction errors compound over the rollout (Bengio et al., 2015; Chen et al., 2024; Huang et al., 2025b): against control50 (16 matched pairs), MLR rises from 30.8% to 41.7% and Gemini-IF falls from 27.08 to 12.50.

Algorithm 7 Causal Frontier training and generation. Future blocks never enter the input; generation commits each block irreversibly.

Input: video of 24 latent frames split into blocks of 4; the first latent frame is the condition.

Training. (i) Sample one frontier uniformly per step. (ii) Feed the committed prefix as clean, detached latents with a near-zero noise level ($\sigma_h=10^{-4}$) and a mask channel set to 1. (iii) Corrupt the active block at noise σ and concatenate prefix and active block as the input; future blocks are excluded. (iv) Apply the denoising loss on the active block only, normalized by the number of active tokens.

Generation. (i) Denoise the next block (30 steps, CFG 7.0) conditioned on the clean prefix. (ii) Commit the block: detach and append it to the prefix. (iii) Repeat until all blocks are generated; there is no overlap or refinement pass.

D.5 ICH-D

ICH-D applies a learned deletion residual during inference. A frozen linear probe estimates the duplicate probability of each latent cell from novelty and consistency features (16-dimensional statistics collected from several Transformer blocks). A gated residual, whose output projection is initialized to zero and trained before inference, then removes the flagged content from the block-22 representation:

$$M = \text{sigmoid}(\beta^\top \phi(h)), \quad h_{22} \leftarrow h_{22} - \text{sigmoid}(g) \cdot M \odot W_{\text{out}}(h_{22}), \quad (24)$$

after which the denoising loop proceeds normally. Over 368 matched pairs against the pretrained model (four seeds), MLR rises from 11.2% to 13.1% and Gemini-IF falls from 45.83 to 31.79; the deletion path introduces its own errors.

Algorithm 8 ICH-D inference-time deletion. The probe β is frozen after offline fitting; W_{out} is initialized to zero, trained beforehand, and frozen at inference.

Input: frozen per-cell probe β , trained gate g , and trained output projection W_{out} .

Output: corrected block-22 representation at each denoising step.

1. At the block-22 output of every denoising forward pass (both CFG branches), compute the per-cell novelty and consistency features ϕ .
 2. Score each cell with the frozen probe, $M = \text{sigmoid}(\beta^\top \phi)$, giving a duplication-probability map over the latent grid.
 3. Erase the flagged content inside the representation via Eq. (24).
 4. Continue the denoising loop normally; there is no re-noising or repair pass.
-

Across the matched comparisons, the five alternatives trade off MLR and instruction following, motivating direct restoration and consistency supervision in the backbone.

E GENERALIZATION RESULTS

E.1 GENERALIZATION ON EWMBENCH

Table 26 reports all EWMBench components and the official Motion, Semantics, and Overall scores. Relative to Standard SFT, EVEWORLD improves Motion by 3.48% and Overall by 1.24%, with a 17.94% gain in DYN. Other components remain close to the matched baseline, with Standard SFT higher on BLEU, Diversity, HSD, and nDTW, and GigaWorld-0 higher on CLIP and Scene. Logic is identical across all three models. The gains of EVEWORLD are concentrated in motion-related and aggregate scores rather than uniformly across all EWMBench components.

E.2 FLOWWAM CHECKPOINT ROBUSTNESS

We evaluate six FlowWAM checkpoints on a fixed RoboTwin slice consisting of episode 45 from each of 50 tasks (Mu et al., 2025). EVEWORLD improves PSNR by 2.20–6.22 dB and increases SSIM at every checkpoint (Table 27), showing consistent reference-fidelity gains across checkpoints.

Table 26: **Complete EWMBench results.** Panel (a) reports semantic and scene components. Panel (b) reports motion components and the official Motion, Semantics, and Overall scores. Scores are rescaled for readability: component and Motion scores are multiplied by 100, and Semantics and Overall by 10.

(a) Semantic and scene components					
Method	BLEU $\uparrow$	CLIP $\uparrow$	Diversity $\uparrow$	Logic $\uparrow$	Scene $\uparrow$
GigaWorld-0	33.12	89.09	2.77	100.00	89.58
Standard SFT	34.44	86.86	3.47	100.00	84.38
EVEWORLD	34.23	87.74	3.26	100.00	86.37

(b) Motion components and aggregate scores						
Method	DYN $\uparrow$	HSD $\uparrow$	nDTW $\uparrow$	Motion $\uparrow$	Semantics $\uparrow$	Overall $\uparrow$
GigaWorld-0	7.19	23.29	19.83	50.30	22.50	36.49
Standard SFT	14.88	24.35	22.28	61.51	22.48	37.07
EVEWORLD	17.55	24.12	21.97	63.65	22.52	37.53

Table 27: **FlowWAM fidelity across checkpoints** on a fixed RoboTwin slice. Each checkpoint is evaluated on episode 45 from each of 50 tasks. Core-8 denotes the mean of eight no-reference metrics from the same evaluator. Δ PSNR reports the gain of EVEWORLD over the control.

Epoch	Control			EVEWORLD			Δ PSNR (dB)
	PSNR (dB) $\uparrow$	SSIM $\uparrow$	Core-8 $\uparrow$	PSNR (dB) $\uparrow$	SSIM $\uparrow$	Core-8 $\uparrow$	
0	16.79	0.807	47.8	18.99	0.857	43.3	+2.20
1	15.38	0.785	47.0	20.23	0.871	48.0	+4.85
2	14.92	0.757	50.5	21.14	0.885	45.9	+6.22
3	15.76	0.785	50.1	18.44	0.839	48.7	+2.68
4	15.02	0.770	48.2	18.02	0.834	47.9	+3.00
5	15.18	0.767	48.5	18.45	0.843	47.8	+3.27

F QUALITATIVE RESULTS

F.1 INSTANCE CONSISTENCY

Figures 11 and 12 provide qualitative examples of target-instance consistency during manipulation. Standard SFT may introduce surplus or missing target instances over the rollout, whereas EVEWORLD better maintains target-instance consistency throughout the interaction.

F.2 CROSS-FRAME CONSISTENCY

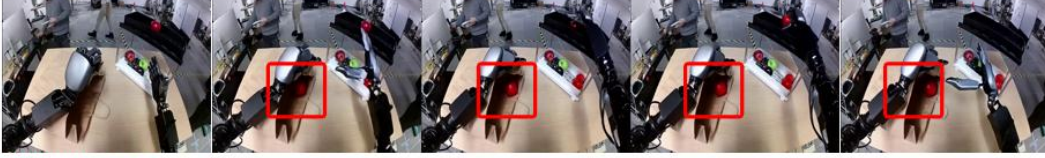
Figures 13 and 14 compare Standard SFT, IGR-only, and the full EVEWORLD at matched stages of the same manipulation. Standard SFT exhibits target duplication during the rollout, and IGR-only can still retain such cross-frame inconsistencies despite directly supervising instance restoration. With TIA added, the full EVEWORLD maintains a single coherent target across the corresponding interaction stages. These examples illustrate the complementary roles of IGR and TIA in preserving target consistency over time.

F.3 TARGET EVOLUTION COMPARISON

MLR intentionally measures changes in target-instance count and therefore does not capture failures that preserve N_0 while corrupting the target itself. For qualitative analysis, we assess such distortion separately using the same VLM-assisted temporal review framework described in Appendix A.6, supplemented by manual review for a subset of cases. A distortion is marked when the target count remains unchanged but the target undergoes an *unsupported* change in shape, appearance, or identity that is inconsistent with its preceding temporal state. Figures 15 to 20 show two representative tasks using the same 24 timestamps for all models. In both examples, Standard SFT loses the target and

thus produces an MLR count violation, whereas IGR-only preserves the count but progressively distorts the target. EVEWORLD preserves both the target count and its cross-frame consistency.

Standard SFT | Model Laziness



EVEWorld (ours)



Task: Use the right hand to pick up red tomato from upper black tray of plastic shelf to inside of brown paper bag.

Standard SFT | Model Laziness



EVEWorld (ours)



Task: Pick up the rounded grey rectangular eraser and place it on the rectangular plastic tray, clearing the rectangular keyboard.

Standard SFT | Model Laziness



EVEWorld (ours)

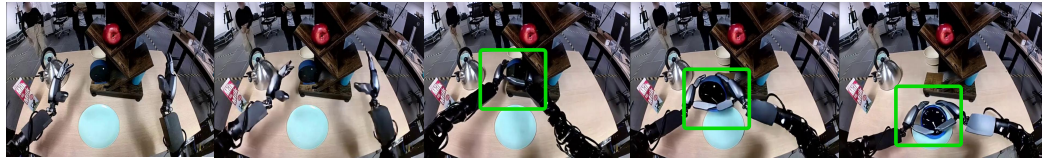


Task: Place the white boxdrink with sturdy lid on the light gray streaked wooden coaster.

Figure 11: **Target-instance consistency under Model Laziness.** Red marks Standard SFT, where Model Laziness violates target-instance consistency during manipulation. Green marks EVEWORLD, which maintains target-instance consistency throughout the interaction.

Standard SFT | Model Laziness**EVEWorld (ours)**

Task: Use the right hand to pick up long-reach lighter to light the candle

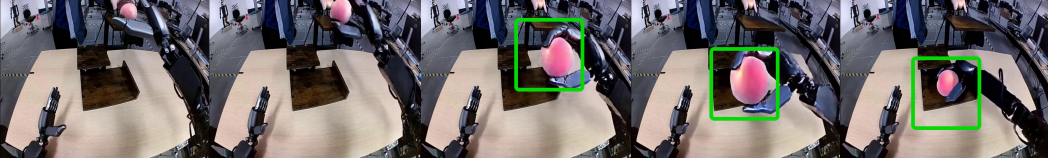
Standard SFT | Model Laziness**EVEWorld (ours)**

Task: Use the left hand to pick up blue clock from bottom wooden shelf to center of teal plate.

Standard SFT | Model Laziness**EVEWorld (ours)**

Task: Use the right hand to pick up plate and place it onto the dish rack

Figure 12: **Additional Model Laziness cases.** Red marks Standard SFT, where target-instance consistency is violated during manipulation. Green marks EVEWORLD, which maintains target-instance consistency throughout the interaction.

Standard SFT | Model Laziness**IGR (Ours): A Direct Approach****EVEWorld (ours)**

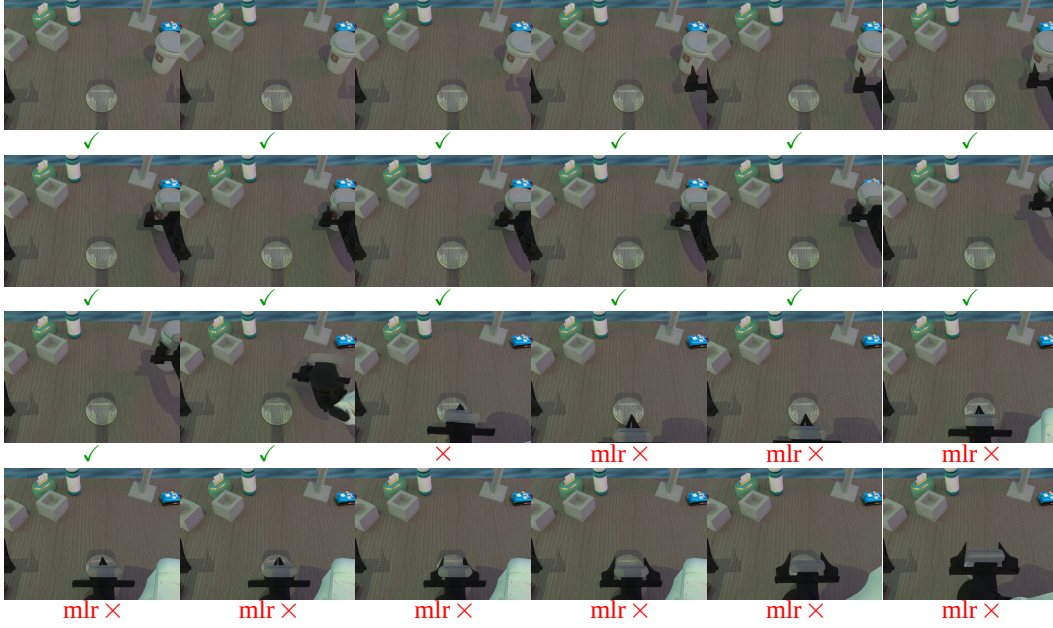
Task: Use the right hand to pick up pink peach from top level of the shelf to bottom level of the shelf.

Figure 13: **Cross-frame target consistency.** Standard SFT and IGR-only show target duplication, appearance drift, or deformation (red), while the full EVEWORLD preserves target identity and appearance across matched interaction stages (green).

Standard SFT | Model Laziness**IGR (Ours): A Direct Approach****EVEWorld (ours)**

Task: Use the right hand to pick up the bread slice from the toaster and place it on the blue plate.

Figure 14: **Cross-frame target consistency.** Standard SFT and IGR-only show target duplication, appearance drift, or deformation (red), while the full EVEWORLD preserves target identity and appearance across matched interaction stages (green).



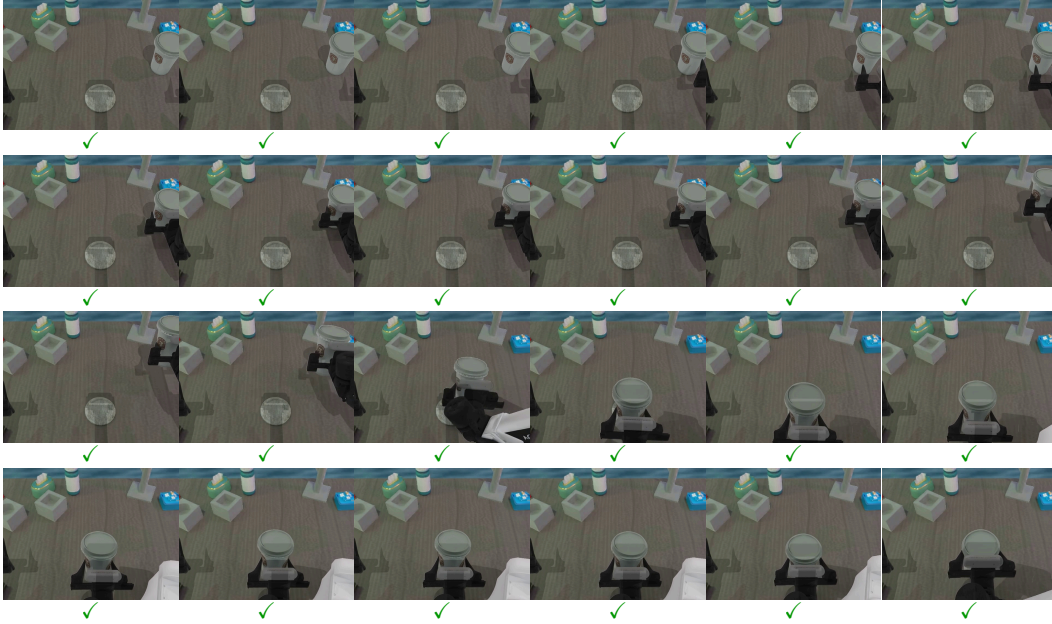
Task: Move the boxed drink the drink box with brown circular print onto the coaster the coaster made of smooth wood.

Figure 15: **Instance inconsistency.** The 24 fixed MLR timestamps are shown in temporal order. **Check marks** denote count-consistent states, while **crosses** denote count deviations. The first cross marks the onset; an MLR event is counted only after two consecutive violating timestamps to suppress isolated detection errors.



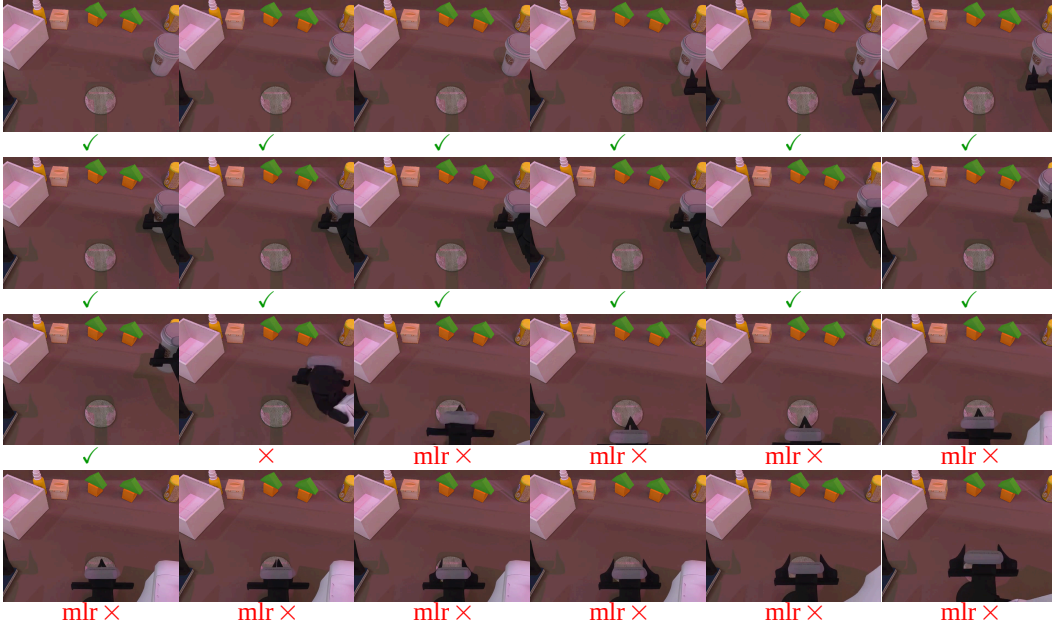
Task: Move the boxed drink the drink box with brown circular print onto the coaster the coaster made of smooth wood.

Figure 16: **Cross-frame inconsistency.** The 24 fixed timestamps are shown in temporal order. **Check marks** denote consistent target states, while **crosses** mark persistent appearance, geometry, or identity changes. The target count is preserved, so these failures are recorded separately from MLR.



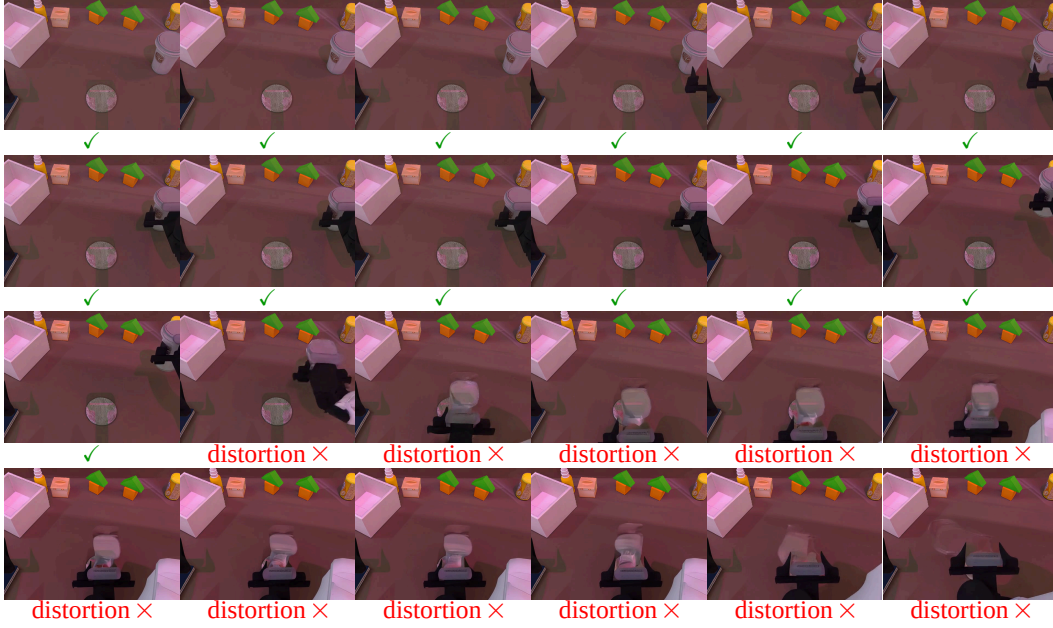
Task: Move the boxed drink the drink box with brown circular print onto the coaster the coaster made of smooth wood.

Figure 17: **Physical consistency.** The 24 fixed timestamps are shown in temporal order. **Check marks** indicate consistent target states throughout the rollout. The target preserves both its instance count and cross-frame appearance, with no persistent count violation or distortion.



Task: Put the drink container with tapered shape squarely on top of the smooth round palm-sized coaster.

Figure 18: **Instance inconsistency.** The 24 fixed MLR timestamps are shown in temporal order. **Check marks** denote count-consistent states, while **crosses** denote count deviations. The first cross marks the onset; an MLR event is counted only after two consecutive violating timestamps to suppress isolated detection errors.



Task: Put the drink container with tapered shape squarely on top of the smooth round palm-sized coaster.

Figure 19: **Cross-frame inconsistency.** The 24 fixed timestamps are shown in temporal order. **Check marks** denote consistent target states, while **crosses** mark persistent appearance, geometry, or identity changes. The target count is preserved, so these failures are recorded separately from MLR.



Task: Put the drink container with tapered shape squarely on top of the smooth round palm-sized coaster.

Figure 20: **Physical consistency.** The 24 fixed timestamps are shown in temporal order. **Check marks** indicate consistent target states throughout the rollout. The target preserves both its instance count and cross-frame appearance, with no persistent count violation or distortion.

G RELATED WORK

We group related work into structured embodied world models and physical alignment, object editing, and approaches based on correspondence, tracking, or memory.

Structured embodied world models and physical alignment.

- **4D scene representations.** Tesseract extends video generation to jointly predict RGB, depth, and surface normals for 4D scene reconstruction (Zhen et al., 2025). While its richer outputs improve geometric modeling, EVEWORLD retains the host model’s native representation and adds post-training restoration supervision for target-instance consistency.
- **Disentangled dynamics.** DVG-WM separates a low-resolution dynamics preview from high-fidelity video refinement and evaluates whether the robot acts on the intended object (Shan et al., 2026). Correct target selection, however, does not ensure that the target remains a single consistent instance throughout the interaction; IGR explicitly trains recovery from count-perturbed demonstrations.
- **Object-aware training.** DreamX-Phi combines action conditioning and depth prediction with target-mask-weighted reconstruction and relational supervision from a frozen V-JEPA teacher (DreamX Team et al., 2026). TIA adds layer-selected local matching and feature transport for the manipulated target across adjacent frames, while IGR supplies an explicit restoration target when instance consistency is violated.
- **Physics-preference learning.** ABot-PhysWorld uses vision-language discriminators to select preferences for physically plausible interaction videos, including contact behavior (Chen et al., 2026b). Its video-level preference signal does not explicitly pair a duplicated target with a clean future; IGR uses that pairing to directly supervise instance-consistent target evolution.

Object editing and corrective supervision.

- **Mask-free video editing.** LoVoRA uses learnable localization to add or remove objects in a supplied video from text, without an external mask at inference (Xiao et al., 2025b). IGR instead uses object insertion as a training perturbation; EVEWORLD predicts the future from an initial image and instruction without an edit request identifying a spurious target.
- **Effect-aware removal.** ROSE removes a specified object and its effects on the scene, including shadows and reflections, from an existing video (Miao et al., 2025). IGR uses the inserted target for a different purpose: restoring the clean manipulation demonstration during post-training, so the world model learns instance consistency before generating future interactions.

Correspondence, tracking, and persistent memory.

- **Joint generation and tracking.** Track4Gen jointly learns video generation and point tracking to reduce appearance drift (Jeong et al., 2025). TIA instead selects a layer using target-matching error and locally matches and transports the manipulated target’s features between adjacent frames, directly supervising its cross-frame consistency during generation.
- **Geometry-distilled correspondence.** GEM-4D distills dense 4D correspondence from a geometry model into the video backbone, improving geometric consistency without an auxiliary branch at inference (Zhou et al., 2026). TIA focuses its matching and transport on the manipulated target, while IGR adds restoration supervision for instance errors that geometric correspondence alone does not explicitly correct.
- **Scene-revisitation memory.** WorldMem stores observations indexed by poses and timestamps to preserve spatial and temporal consistency when a simulated agent revisits a scene (Xiao et al., 2025a). Our evolution supervision instead addresses errors in how the manipulated target changes during an interaction, including unsupported duplication and cross-frame drift.
- **Object-centric memory.** SlotMemory organizes key-value features around reusable object slots, improving long-video entity persistence and showing qualitative object-count stability (Dou et al., 2026). IGR complements such persistence by explicitly restoring count-perturbed manipulation videos to supervise instance consistency, rather than relying on stored entity features alone.
- **Manipulation history retrieval.** Mem-World uses a 4D surfel memory and action-conditioned history retrieval to maintain scene information under wrist-camera motion and occlusion (Zheng

et al., 2026). This historical context complements our direct supervision of the manipulated target: IGR promotes instance consistency, and TIA promotes cross-frame consistency during video prediction.